

Zero Day Malware Detection in the Windows Ecosystem : A Hybrid Contrastive and Behavioral Learning Approach

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A thesis submitted to the Department of Computer Science and
Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science and Engineering

Department of Computer Science and Engineering
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June 2025

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Declaration

It is hereby declared that

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3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
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Abstract

The fast development of digitalization of the basic activities of the every-day life in this century has conditioned more dependence on the technological devices which made the observable enhancement of interdependence between the ecosystems, especially in Windows . With key areas of services, including banking, medical, and e-commerce services shifting to be online, the insecurities in these systems mean that more cyber jeopardy are faced by users, especially the zero-day malware attacks. Such attacks are also a huge challenge to cybersecurity on desktop and mobile platforms since they take advantage of vulnerabilities that cannot be easily identified and prevented. This thesis proposes a new hybrid method of detecting zero-day malware where ideas of contrastive and behavioral learning frameworks are combined. Through combination of these techniques, the proposed design will enhance the process of detecting and preventing unknown malware in windows platform. The hybrid detection system is able to sample the behavior of applications and compare against unique attributes, and provide a more flexible and effective barrier to defense. This method aims at giving input to the creation of scalable, real-time and Windows-centric platform security-solutions to address the shifting security requirements of the digital ecosystems. This paper proposes a staged framework for malware detection that increases its level of analysis gradually, from static filtering to behavioral modeling, self-supervised byte-level anomaly detection and network-wide monitoring, all within a single pipeline focusing on the Windows platform. Unlike other methods that rely on labeled malware datasets or a one-stage approach, this one focuses on a self-supervised approach and gradual risk assessment to effectively detect previously unseen malware, also known as zero-day malware which is the key novelty of this work.

Keywords: Zero-Day Malware, Windows-centric Platform Security, Hybrid method, Contrastive & Behavioral learning, Real-time detection, Digital ecosystems

Acknowledgement

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Chapter 1

Introduction

1.1 Background

A zero-day exploit is a cyberattack that uses an uncovered or unfixed security vulnerability in computer hardware, software, or firmware [5]. The term "zero-day" refers to a software or hardware fault that a vendor cannot repair before hostile actors exploit it to get access to affected systems [31]. These vulnerabilities are particularly dangerous because they can be exploited before any defensive measures—such as patches or updates—are implemented, leaving systems open to attack. These exploits can lead to unauthorized access, data breaches, and long-term malware deployment [33]. Attackers use a variety of channels to exploit zero-day defects, including user interaction and network connectivity. One popular route is phishing, which involves deceptive emails or messages tricking users into clicking corrupted links or downloading files that activate the zero-day vulnerability. Network-based attacks analyze systems for unpatched vulnerabilities or open services before sending malicious material to exploit them remotely. Infected portable media, such as USB drives, can also spread malware by exploiting a vulnerability when they connect to a computer. Real-world incidents like the Nimda worm, which rapidly spread through email, network shares, and web server vulnerabilities, and the SQL Slammer worm, which caused widespread denial-of-service by exploiting a SQL server vulnerability, demonstrate the devastating speed and reach of zero-day attacks when combined with multiple propagation vectors[3].

The diagram below presents the life cycle of zero-day vulnerability-

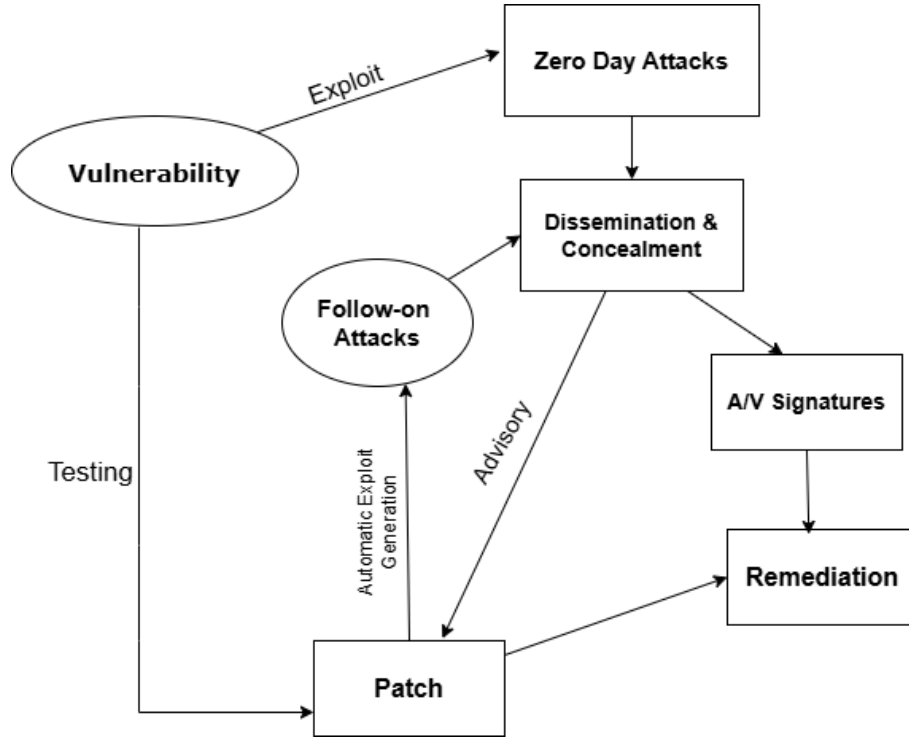


Figure 1.1: Zero-day Vulnerability Life Cycle [3]

As shown in figure 1.1, the life cycle of a zero-day vulnerability consists of seven stages. Zero-day vulnerabilities typically start out as software bugs. If the software maker doesn't find the problem, attackers could be able to exploit it. When software makers find a vulnerability, they usually make the security issue public [9]. The vendors then update anti-virus signatures to reflect the new understanding of zero-day assaults and release a patch for the zero-day vulnerability.

Even after a vendor has detected and addressed a vulnerability, fresh exploits may be developed and deployed against targets who have not yet received the patch. It may take several years for remediation to occur following this cycle of patching and exploitation, at which point the vulnerability no longer affects systems [16]. Despite growing awareness and improvements in cybersecurity solutions, the detection and prevention of zero-day attacks remain profoundly challenging. Mainly because of their unknown nature, polymorphism, obfuscation, runtime evasion, and encrypted payloads. These strategies enable malware to conceal its true behavior, resemble legitimate software, and avoid static and dynamic analysis tools. Furthermore, the absence of labeled datasets and high false positive rates reduce the efficacy of machine learning-based detection, while device limitations restrict real-time protection.[22] Solving this problem is not that straightforward, because traditional machine learning methods work well against known threats but struggle when it's unknown. These models mostly rely on previously identified attack patterns because they are usually trained on static, labeled datasets. As a result they have difficulty adapting to unknown threats. For example, if a malware variant uses a slightly different technique, the model will fail to recognize it. The lack of runtime behavioral context, reliance on static features (such as permissions or API calls), and inability to evaluate encrypted or obfuscated payloads all reduce efficiency. Furthermore, imbalanced datasets, in which benign samples outweigh malicious ones, increases the

likelihood of false negatives. This is why a hybrid approach is needed to tackle these limitations. Our plan is to incorporate both behavioral and contrastive learning to get a greater accuracy in preventing and detecting zero-day attacks.

1.2 Rational of the Study or Motivation

The purpose of this research emerges from the growing threat that zero-day malware poses due to its ability to exploit systems and breach data before any defense mechanism is created. As mentioned before that traditional machine learning models often fail to detect these attacks, it has caused significant damage to data security. Moreover, due to its increasing complexity and frequency, these losses are not only limited to data breach but also include grave financial losses.

For example, The HAFNIUM attack was a significant cyber operation attributed to the state-sponsored group HAFNIUM, operating out of China, targeting Microsoft Exchange Servers. In early January 2021, suspicious activities were detected on numerous Microsoft Exchange servers, which upon investigation, were found to involve zero-day vulnerabilities exploited by HAFNIUM to gain unauthorized access. Over 30,000 organizations in the US alone were affected initially, with hundreds of thousands globally impacted, including government agencies, healthcare, law firms, and research institutions. The attack network was exploited to spread ransomware such as Dear Cry, demanding ransom payments up to \$16,000, leading to financial and operational damage .

Another example is, In 2010, a highly advanced computer worm called Stuxnet made headlines when it targeted Iran’s nuclear facilities. It exploited four unknown security flaws in Microsoft Windows [33], allowing it to infiltrate the systems controlling the facility. Once inside, Stuxnet secretly sent destructive commands to the centrifuges used for enriching uranium, making them spin uncontrollably fast until they broke down. In total, about 1,000 centrifuges were damaged in the attack. Android systems are also vulnerable to zero-day attacks mainly due to their fragmented ecosystem. Many reasons like inconsistent software updates, multiple manufacturers with varying android versions and the open source nature of android allows attackers to exploit it easily. There’s also the risk of installing compromised apps, which will later ask for unnecessary permissions. Therefore, there is a significant need for a real time and adaptive detection system that will tackle these problems going beyond traditional approaches. So our proposed solution is a hybrid approach that combines contrastive learning and behavioral analysis allowing for robust detection of zero-day threats, even in restricted environments.

1.3 Problem Statement

Despite making consistent progress on detecting vulnerabilities, zero-day attacks on Windows OS remain a challenge due to its unpredictable nature of attacks and changeability. These attacks are not yet discovered by vendors or developers as they never ceased to exist in the first place. One of the most noteworthy instances of

such an attack is the Eternal Blue exploit which was developed by the NSA and later hacked by a group of threat actors called Shadow Brokers in 2017. Eternal Blue exploit helped the hackers acquire unauthorized access and execute malicious commands in a remote manner along with initiating expeditious spreading of the ransomware across 200,000 computers in 150 countries, which caused approximately \$8 billion impact on the global economic landscape [23].

This poses a significant threat for the scenery of the Windows ecosystem as it is the most popular and widely used operating system because of its user-friendly and intuitive GUI, software compatibility, and application ecosystem. Businesses and individuals, who operate on Windows on a regular basis, are at constant risk of cyberattacks because of these unknown vulnerabilities, which security vendors have not yet identified. The most concerning issue remains with the fact that conventional security measures lack the ability to timely detect and neutralize these new threats. These signature-based techniques prevail as ineffective in detecting zero-day vulnerabilities as these threats are not yet documented or analyzed [19]. Researchers, in recent times, used signature-based detection techniques which enormously depend on identification of malware using formerly known patterns. Such signature-based solutions prove to be insufficient in detecting zero-day patterns as datasets for Machine Learning models have no labeled examples of these attacks. The detection becomes more challenging and strenuous due to the constant altering nature of malware programs such as, obfuscation of codes and polymorphism [6]. As a result, these conventional approaches cannot offer effective real-time detection and prevention of malware; thus, systems remain vulnerable to exploits.

A close challenge is also seen in mobile environments such as Android, which is a frequent target due to its widespread adoption and open-source ecosystem [39]. previous works have shown that permission can be effective for detecting malware in Android ecosystems [11]. Due to platform fragmentation and delayed security patch deployment, Android shows additional complexities that are beyond the scope of this work. Therefore, this research focuses exclusively on zero-day malware detection within the Windows operating system.

Although attempts are made by several researchers to reach a solution to discover and diminish real-time zero-day attacks facilitated by unknown vulnerabilities using Machine Learning models and heuristic analysis, the methods were not totally efficient as the procedures could not effectively dispense and deal with the newly sophisticated version of these attacks [2] [19].

Hence, this study's objective is to establish a hybrid approach by combining **Constrastive** and **Behavioral** learning techniques with adaptive learning feedback loop to systematically discover and mitigate zero-day attacks, designated specifically for Windows ecosystems. The purpose of this line of action is to refine zero-day vulnerability detection accuracy and efficiency which will minimize the chances of exposure to new and advanced malware. In this manner, it will help create a safer Windows ecosystem with vulnerabilities mitigated before they can be used extensively by attackers.

1.4 Research Questions

RQ1: How effective is a layered escalation-based framework in detecting zero-day malware within the Windows ecosystem compared to single-stage detection approaches?

RQ2: Can self-supervised byte-level anomaly detection identify previously unseen malware without relying on labeled malware datasets?

RQ3: How does combining static analysis, behavioral monitoring and network-level observation improve zero-day detection accuracy while controlling false positives?

RQ4: What is the impact of staged escalation on system performance and detection efficiency in real-time Windows environments?

1.5 Objective

In recent years, the increasing sophistication of cyberattacks, particularly zero-day threats, has highlighted the limitations of traditional signature-based detection systems. As attackers exploit unknown vulnerabilities, there is a growing need for adaptive security solutions. This research addresses this gap by leveraging artificial intelligence (AI) for detecting malware through **behavioral analysis**.

Key Objectives

1. **Establish Behavioral Baselines for Common File Types:** Develop baseline behaviors for file types like RAR files, focusing on their typical interactions with system resources.
2. **Detect Deviations from Expected Behavior:** Flag atypical actions, such as a RAR file attempting to launch the command prompt, access the registry, or initiate network connections—actions that could indicate malicious behavior.
3. **Implement AI and Machine Learning Models:** Develop models that learn from both malicious and benign activity patterns, improving detection accuracy over time.
4. **Create a Real-Time Response Mechanism:** Build a system that can automatically block suspicious files before they execute harmful actions.
5. **Minimize False Positives and Enhance Adaptability:** Focus on reducing false positives and making the system adaptable to evolving threats, ensuring continuous protection.

By integrating behavioral analysis and AI-based decision-making, this research aims to create a proactive, intelligent security system that outperforms traditional signature-based methods in detecting zero-day threats.

1.6 Methodology in Brief

This research proposes a tiered malware detection system that is able to identify known malware as well as unknown malware, referred to as zero-day malware, that targets Windows-based systems. The system is based on a tiered model in which files pass through a series of detection levels, with further analysis applied to those that are initially suspicious. The intention is to improve detection rates without increasing false positives or the processing overhead.

The system has four different layers that gradually build on one another. Each layer adds another perspective on the file. The first layer is a fast static baseline that rapidly scans through all the known threats with fast initial filtering. It essentially checks the file's metadata and structural properties that are commonly associated with previous malware. This layer eliminates harmless files with speed[1].

The second layer involves behavioral modeling. This layer analyzes the file based on the behavior of the file when executed. It also involves rule-based analysis. This layer checks if the file behaves in a manner that is consistent with the type of file it is. For example, a document file should not trigger system actions. This layer identifies suspicious behaviors using rule-based analysis[3].

The third layer involves in-depth analysis of the file contents using self-supervised learning. This layer analyzes the file's contents at the byte level. Unlike the previous layer, this layer does not look at the file's structure; instead, it learns what normal and benign files look like. This layer is very good at detecting zero-day malware [8]. The last layer involves combining the results of the previous layers. This layer combines the anomaly scores of the previous layers and the escalation rules. This layer generates a final decision on whether the file should be allowed, escalated for further analysis, or blocked. Although the system also has the capability to escalate the file for further analysis, the main aim of this layer is malware detection[36].

Experiments have shown that the system improves the effectiveness of malware detection, including the zero day malware detection[22].

1.7 Scopes and Challenges

Scope of Hybrid Approaches in Detecting Zero-Day Attacks

Traditional malware detection models mostly fail to detect zero-day attacks because those models mostly work for known malware. In the hybrid model, we are combining multiple detection approaches together which is increasing the efficiency of the model in detecting zero-day attack. This approach covers a diverse analysis approach for different types of files [31] [30].

1. Comprehensive Threat Coverage

Hybrid model is a combination of static analysis, behavior monitoring and also machine learning approaches. This increases the chance of detecting known and unknown malware. It can also detect hidden malware within any kind of file type. This model overcomes the limitation of other approaches because it adds multiple

layers which helps in increasing the efficiency. Some models work well for executable files but fail to detect hidden malware. This model overcomes those shortcomings [22].

2. Real-Time Threat Response and Containment

This hybrid model has software defined networking (SDN) in layer three which isolates the infected files or suspicious files. This real time working feature helps to minimize the damage if the system gets infected and also helps to prevent malware from working in the system [36].

3. Scalability for Enterprise Environments

This model can be deployed into multiple environments and systems. By federated learning, we can collaborate between different systems by sharing learned threats while not exposing any kind of user data. This approach will help to respond faster and also it will make the model more accurate and updated [31].

4. Adaptability Through Feedback Loops

This model learns and updates itself with the feedback in multiple layers for a file. Behavioral analysis can update the signature based attack database with new detected attack data. New unpacked malware patterns can update the static analysis database. This approach will help the model to be relevant with time and it will be also known to newer malware and how they approach in a system [31].

Challenges in Detecting Zero-Day Malware Using Hybrid Approaches

1. High Computational and Resource Overhead

Omnipack approach needs real time execution tracking because it monitors the activities of secondary memory which requires a lot of CPU and primary memory. CNN or VGG16, these deep learning models need a strong GPU to work faster. It is hard to use these models in low devices. These systems may get slower if we use them to store a lot of files [3].

2. False Alarms and Confusing Behaviors

Sometimes normal program behavior looks suspicious like changing their own code in the memory, this thing happens when an app gets updated. Some advanced apps behavior can be confusing like word files. They can be flagged without being a malware. It's actually hard to maintain a perfect behavioral analysis for each file so that safe programs do not get flagged [20].

3. Difficulty Keeping Rules Up to Date

YARA patterns or PE file structures need to be updated because malware keeps changing with time. Threat actors often use file-type spoofing to hide the malware in different file types. Rules need to be checked and updated with newer file types and how newer malware behavior is changing [6].

4. Machine Learning Can Be Tricked

Hackers can create files which are a little bit different from known malware. It can confuse the ML models into thinking the files are safe. Malware can be encrypted which makes them hard to recognize by the ML model. Without a proper approach with the ML models, it can be hard to detect zero-day attacks [22].

5. Malware That Hides or Avoids Detection

Zero-day malware always use newer approaches to make them undetectable. If they detect that they are executing in the sandbox, the malware can hold their running. Malware sometimes runs child processes so that they can hide them from getting flagged. Malware can also be injected into trusted sources so that they can gain user trust even if the models flagged them. To detect these kinds of malware tricks, it often needs the model's deep access in the system which can create privacy issues [20].

6. No Standard Way to Combine Techniques

There is no method which can mix static analysis, behavioral analysis and machine learning together. So, these methods need different custom setup and also need to be connected with each other in order to work together as a different layer of a single system. They will not work together without any kind of communication link between them [31].

7. Not Enough Good Data for Training

There are many malware datasets but those are obsolete or incomplete. Real newer malware samples are hard to find because of evasion and obfuscation techniques. If we train our machine learning models with old datasets, they would not be particularly effective at detecting zero-day threats [6].

Chapter 2

Literature Review

2.1 Preliminaries

Zero-day malware is the type of malware that exploits the vulnerabilities no one knew about until after the damage starts. So spotting it with traditional antivirus is extremely difficult. As there is no defence ready when the attack takes place, these threats bypass the standard security protocol. With the increasing complexity and diversity of modern devices, zero day attacks are no longer limited to a single platform. Windows with its vast personal and professional use remains the prime target of these attacks. Also , android - the most popular mobile operating system is also exposed due to their open environment and vast app store. Most existing detection methods that scan codes statically or rely heavily on well known behavioral patterns- they fail to discover novel or complex polymorphic malware. This research proposes to address these limitations with an hybrid approach that incorporates behavioral analysis with contrastive learning for zero-day malware detection on Windows ecosystem. The aim is to build a robust, windows-focused detection model that can spot brand-new threats it has never met before. Behavioral analysis watches how the code acts while in execution and contrastive learning improves the system's ability to tell apart subtle similarities and glaring differences between harmless and harmful behavior. Combining these two approaches should boost both the accuracy and the real-world flexibility of zero-day malware detection targeting Windows systems.

2.2 Review of Existing Research

This paper highlights the limitation of existing IDS solutions that primarily focus on known attack signatures or outdated datasets. Traditional IDS methods often fail to detect these zero day attacks as they rely on predefined attack signatures or outdated datasets, which do not account for evolving attack strategies. The paper also discusses the limitations of training models on outdated or biased datasets and the importance of retraining models to adapt to new attack types. Though deep learning models show promise in novelty detection, factors like limited memory, computational demands and deployment settings can impact model performance in real-world scenarios. So, the review calls for exploring various techniques and

algorithms for zero-day attack detection and keeping an eye on emerging trends and hurdles in network anomaly detection. It also emphasizes the importance of developing IDS systems that are robust, unbiased, and capable of detecting both known and unknown attacks.

Huthifh et al. [12] develops a new zero-day attack prevention and detection mechanism for Software-Defined Networks (SDNs) by customizing the Cuckoo sandbox tool. A sandbox environment is used to execute untested programs in isolation to prevent infection of real environments. The modified Cuckoo sandbox enhances zero-day attack detection by providing an isolated and automated environment where unknown files or URLs can be executed safely for analysis. During this process, it profiles the behavior of potential malware across multiple Windows VM environments, capturing memory dumps, registry changes, and created files. This behavior-based analysis allows the system to generate a severity score for each threat, facilitating accurate identification of malicious activity even for previously unseen vulnerabilities. Furthermore, integrating the sandbox within the SDN controller enables real-time monitoring and immediate response, such as detaching infected clients or restricting malicious traffic, thus preventing the spread of zero-day malware. Cuckoo Sandbox successfully identified 353 out of 361 tested malwares, achieving a success rate of 97.78%. But it doesn't extensively address scalability, real-world deployment challenges or evasive malware techniques. Future work could involve integrating machine learning for better detection, validating in diverse environments, and developing evasion-resistant sandboxing methods.

There's a case study on HAFNIUM Attack in this article. The HAFNIUM attack targeted Microsoft Exchange servers using multiple zero-day vulnerabilities. The attackers, believed to be a state-sponsored organization executing from China, compromised a substantial number of organizations globally, with estimates indicating that over 30,000 organizations in the US and hundreds of thousands worldwide were affected. This included a variety of industries, such as law, healthcare, education, and defense. To counter these threats, the article emphasizes the importance of preemptive defence strategies, including rapid patch release, advanced intrusion detection and prevention systems, behavioral analytics, and network segmentation. Maintaining an up-to-date inventory of software libraries, automating patch management, and fostering a security-aware organizational culture are critical steps in reducing vulnerabilities. However, they have some limitations. They often lack detailed insights into real-time detection techniques for zero-day exploits, comprehensive methodologies for proactive threat hunting, and practical implementation challenges of suggested solutions. Additionally, future work should focus on developing automated, adaptive security frameworks to improve early detection and prevention of zero-day attacks

This study [36] presents ZSDN-Guard, a Zero Trust-based protection framework for SDN, which uses deep learning for live anomaly detection and real-time access regulation. Existing SDN security models rely on machine learning-based anomaly detection but often use outdated datasets. The problem with this current method is that most methods focus on securing the controller rather than protecting the entire network. Perimeter-based defenses cannot adapt to dynamic threats in SDN envi-

ronments. So this work proposes various solutions like ZSDN-Guard or NetSeqDL (a deep learning model) to detect network anomalies with 99.75% accuracy. As well as implementing dynamic trust-based access control instead of static rules. It highlights that future work will focus on further improving the framework's performance, particularly in more complex network environments, and exploring its applicability across diverse and large-scale networks.

Ibraheem & Tosho [27] presents that traditional security methods rely on signature-based detection, which is ineffective against zero-day attacks. Recent research explores machine learning models for live abnormality monitoring, with random forests, decision trees, SVM, and gradient boosting emerging as promising techniques. But ML-based approaches face challenges like data collection, labeling, and integration into security workflows. The study proposes using machine learning models, particularly random forest, to detect zero-day attacks based on data pattern analysis. The models were trained and verified on CSE-CIC-IDS2018, ensuring accurate detection of attack patterns. The results indicate random forest's superior accuracy, making it a strong candidate for improving cybersecurity defenses.

The paper surveys the current state of zero-day attack detection and protection, highlighting that signature-based methods, while effective against known threats, are inherently reactive and fail to address new, unknown vulnerabilities in real-time, leaving systems vulnerable during the window before patches are available. It discusses techniques such as protocol behavior analysis, anomaly detection, data matching, honeypots, traffic signatures, vulnerability-based signatures and innovative approaches like using C# for automatic worm detection. The primary challenge with existing methods is their inability to promptly identify and prevent zero-day attacks due to rapid threat evolution and the limitations of reactive signature updates, which can take hours or days, during which attackers can exploit unpatched vulnerabilities. Future directions suggested include developing proactive, AI-driven detection mechanisms, enhancing behavioral analysis, automating worm detection with intelligent systems, and integrating multi-layered security architectures to provide higher performance and more robust zero-day defenses, thereby reducing the window of vulnerability.

The title of this paper is "Zero-day attack detection and prevention in software defined networks". This approach works using the SDN environment. In zero-day attack, the attacker exploits an unknown issue of a system. Mostly, traditional ways to detect malware. The authors of this paper proposed a dual layered defence system using cuckoo sandbox. All traffics which are passing through the switches is inspected by SDN controller, it extracts files and URLs and analysis them in a windows based sandbox. The files are assigned a number by the sandbox indicating how well the files behave with malware. A custom UNIX-based sandbox that monitors the critical components which is used for the controller protection. It monitors the service ports and OS features which detects any kind of unauthorized access and triggers the alarm. This paper's approach is done by using Mininet and Openflow. They used malware of size from 2 KB to 1400 KB and got detection rate of 97.78%. This shows the effectiveness of this approach. In future, we can try to make this approach better by increasing the malware size [13].

This paper shows a deep learning approach for malware classification where they used pre-trained VGG16 CNN model and it also shows us the challenges of traditional signature based methods. The researchers converted a malware into a grayscale image and each malware creates a different image. The pre-trained image recognition system is used to analyze those images. The system was 98% accurate in detecting malware types[2]. This system also works for most of the kind of malware. This approach is faster than trying a new system because they used a pre-trained system and made it better. But there are also some issues like some malware type was over presented in the test data. This system struggles a lot in detecting malware with mostly same visual variants. Advanced malware which hides its code can bypass this system because it only looks for the code. This system also should look for the behaviours of a file which can increase its accuracy to detect zero-day attack. For future research, this researchers suggested that we should add a way to find out unknown patterns of a malware and also we checkout how a program is behaving when we are running it [14].

A malware always changes its code to avoid detection. This paper tried to solve this issue. The system they developed mainly checks how files call system functions as well as code pattern. They tried three different machine learning ways. The methods used are Decision Tree, Random Forest and LightGBM. Within all of them lightBGM worked best because it made less mistakes then others. But sometimes it also missed some actual malware which is very risky for a system. This research suggest us that we need bigger datasets to know about how newer malware behaves while running. This system normally detects normal known malware but it also can detect unknown malware by checking that if its pattern matches with the pattern of a malware file. This system mainly check out the byte sequence, opcodes, PE header and API calls from malware file, it helps to detect a malware without executing the file [14]. But to make this system more accurate they needed bigger datasets which is a limitation of them. False alert was the biggest issue in this system. To make this system better they could try a hybrid approach by combining multiple systems [14] [22].

Omnipack is a way to detect a malware when the malware is compressed or encrypted so that it can bypass normal signature based malware detection systems. Omnipack checkout that how a programme is getting executed. It uses hardware based assisted memory protection. This tracks that when a program is executing its code from the memory page its previously used, it's a sign of that other code is getting executed from that place after execution of the main code. Omnipack has high efficiency because it has hardware feature like non-executable page and using Coarse-grained monitoring. It also increases the safety by using malware detectors like clamAV only when potentially harmful system calls like read ,write are tried,it checks that if those calls are safe before they can harm the system. This approach is practical because it takes different amount of times for detecting packed and unpacked malware. This approach performs better than other ways of preventing malware [1]. It also has issues such as giving false alarms because it cannot monitor the child processes. Despite having some issue, this approach is a production ready solution for the malware detection system [1] [22].

JothiShri et al. [28] consider the issue of deep learning in network security, paying attention to the problem of anomaly detection and real time threat analysis. CNNs and RNNs are used in the study to boost cyber security defences. Namely, CNNs are used on network traffic data and their capacity is to extract spatial patterns in packets. By contrast, RNNs are used to perform time series processing over network-based log data identifying bad trends over time. CNNs are aimed at identifying the deviations that can be perceived as possible breaches, whereas the RNNs are geared at finding the anomalies like the unauthorized logins or the abnormal data transfer. Methodology mentions the hybrid combination of these models, but allows to cover the whole possible set of security incidents. Important indicators of this paper imply that the detection rate used in this research in anomaly detection is 95% with a known attack accuracy of 98% and a prior unknown attack accuracy of 85%. The system shows astonishing performance of 0.8 ms per packet, which indicates how effective the deep learning models are in real-time processes. The paper focuses more on the issue of model interpretability that is of high concern to security analysts who are required to trust AI-driven systems. Irrespective of these limitations, AI models can be extremely useful in the identification of zero-day vulnerabilities owing to their flexibility and adaptability to new and changing information.

Another solution inspired by the work of Sarker et al. [18] introduces a hybrid AI-based model combining both supervised and unsupervised learning architecture that can be used to detect known threats and novel or unknown threats on real-time network traffic via decision trees and k-means clustering. The hybrid model is proposed to offer an assignable solution to the predefined cyber-threats as well as the emerging ones, relying on the two strengths of the particular machine learning techniques. In supervised learning, decision trees are used on labeled data, to detect common patterns of attacks, and this is unlike the unsupervised learning where the use of k-means clustering helps in detecting the undesired behaviors that are outside the current profiled threats. Such hybridization allows making the model ready to pursue a great variety of cybersecurity situations. The authors also use the Natural Language Processing (NLP) to process the unstructured data like the logs and the incident reports and this aspect makes the model able to identify the emerging threat, which is yet not deemed in the structured data. It is a particularly valuable part of NLP when it comes to responding to the incident in real-time, as it allows identifying the threat and its mitigation faster. The measure of performance of the model can be mentioned as 90% accuracy in detecting network anomalies and 92% precision in detecting suspicious behaviors; this indicates that the model is reliable. Nevertheless, its recall in the detection of new and unseen threats is 78%, which means that the model is less efficient when it comes to discovering completely new attack strategies. Nevertheless, the system has a great potential, and its speed is rapid enough to do real-time analysis.

Li et al. [29] presents the PAFE methodology, which is a new framework of the malware naming technique, based on visualization to represent the malware binary as a gray-scale image. This approach does not rely on the conventional possibility of previous domain experience and long periods of execution such as complicated manual feature extraction. The system assists in stopping malware since the binary data in the form of images makes the analysis process simpler and, at the same

time, enables deep learning models to automate feature extraction. This can be especially useful towards large-scale detection of malware, as it is a major concern of cybersecurity in real life. To cope with a usual problem of image distortion when the image normalization process occurs, the authors [21] present a method of padding pixels. Normalization process as the bilinear interpolation sometimes corrupts some fine details of an image, which might lead to the loss of important detail and influence the accuracy of classification. The available pixel-padding technique guarantees that the integrity of the image is not lost as the image is filled with pixels without distorting the main characteristics of the image. Such an approach preserves the key patterns of the texture, that is vital in the differentiations of various malware types and greatly enhances model resilience. This classification model based on the PAFE structure provides high accuracy with 99.25% and a prediction process time of 10.04 ms/malware, which can be regarded as a very high value compared to traditional models, such as VGG16 and ResNet. Although they are accurate, such models are computationally high priced and slow to process, particularly in real time operations. Through the lighter structure approach, PAFE combines the excellent performance and guarantees a high rate of processing speed, making it appropriate in real-time malware diagnosis where speed plays the vital role. when the low prediction time is observed at 10.04ms, it will be able to process large network traffic with it all coming out accurately with hence causing no latency.

Mahindru et al. [11] introduce MLDroid, a machine learning-based, dynamic Android malware detection framework that utilizes web-based frameworks. The framework is aimed at studying the behavior of Android applications during runtime, with the permission and API calls as the main features of discovering malicious behavior. The features are isolated during the execution process and allow for dynamic observation of how the app communicates with the device, something that is so crucial in spotting suspicious behavior, which may be overlooked by static analysis. In this work, the authors employ feature selection methods such as Gain-Ratio, Chi-squared tests and Principal Component Analysis (PCA) which help to refine the input data structure by identifying the most pertinent features to train the detection model. A range of Machine Learning methods, such as deep learning, farthest-first clustering, Y-MLP and a nonlinear ensemble decision tree forest technique, are then trained using the selected features, enhancing the classification efficacy of both harmful and benign apps. With a 98% detection rate, the architecture demonstrates its proficiency by detecting malware with exceptional precision in both known and unknown malware categories. Additionally, by focusing on the most significant features, Rough Set Analysis (RSA) serves to decrease feature redundancy while improving model performance. Despite its superiority at malware identification compared to other machine learning-based malware detectors, the performance of MLDroid is enhanced further by combining two kinds of learning, namely, supervised and unsupervised learning, enabling it to respond to new and unknown threats. Such versatility is especially significant for Android malware, which continuously changes to evade detection.

In their study, Afanian et al. [6] explore the methods applied by malware in order to evade the standard dynamic analysis tools used to track the activity of applications

during runtime. The paper establishes that these evasion strategies are grouped into detection-dependent or detection-independent categories. Detection-dependent Malware is designed with detection in mind, by detecting the presence of an analysis environment e.g., sandboxes or debuggers and modifying its actions to evade detection. NGlobalFlags, for example, is an anti-debugging method, making analysis difficult as it verifies the presence of debugging tools and adjusts its actions accordingly. This poses a risk to traditional manual dynamic analysis as these evasion techniques target analysts employing debuggers to observe malware behavior. Comparatively, detection-independent methods involve malware that does not seek to recognize the analysis environment. These methods are more difficult to notice since they do not modify the external environment but rather the malware’s internal form or behavior, such as fileless malware, which executes directly from memory and is often overlooked by conventional methods. The authors from [6] reveal that manual dynamic analysis only detected 87% of malware and the detection rate sharply declined to only 65% when facing advanced evasion techniques such as anti-debugging or fileless malware. The study also explores the performance of automated sandboxes that have a 90% success rate with known malware but fares poorly with novel versions or obfuscated programs. This shows the shortcoming of the sandboxing technique alone. To enhance the detection, the authors suggest using dynamic analysis methods alongside machine learning and behavioral analysis, as these methods can adjust to the newly found evasion methods more efficiently and provide greater insight into the malware.

Mobile phones have evolved very fast to include smartphones which are a fundamental component of life today and the android version is on the frontline because of its open-source attribute and huge marketplace. This popularity has however made Android, the target of malware developers. The existing traditional antivirus mitigations, which mainly use known malware signatures to detect malware, have not been able to cope with the rise in the number of applications that are being uploaded on such sites as the Android market. According to Grace et al. [2], manual vetting is an impossible feat considering that an average of 1,242 new applications are released each day. The also propose RiskRanker, the proactive system which is able to detect zero-day Android malware based on the analysis of the app behavior, and not on the existing malware samples. In contrast to traditional ways based on signature matching, RiskRanker utilizes the two-order risk analysis so that it can identify the apps that are of high and medium risk. The first-order analysis involves locating behaviors which include root exploits or unauthorized sending of SMS in the background, and flags apps as high risk. In the meantime, the medium-risk apps may not use platform vulnerabilities but can be harmful to a degree that leads to monetary or personal loss, such as unlawfully subscribing the person to services. As Grace et al. point out, RiskRanker has detected 718 malicious apps across 118,318 sampled apps (where 322 were zero-day malware out of 11 malware families). The second-order analysis, that is aimed at more complex evasion methods such as dynamic code loading and encrypted execution of the native code, also optimizes the detection process, managing to spot the presence of more threats. The authors validate that RiskRanker is scalable since it was able to analyze more than 118,000 apps within a span of less than four days, which is unachievable using traditional malware detection systems. Evaluation demonstrated that the proactive nature of

RiskRanker could identify threatening events unknown to the organization, that is, 322 zero-day malware samples. Such an outcome is essential, because conventional systems have the tendency to miss most of the newly emerging threats which are found to exploit vulnerabilities that had not been recorded before. Grace and others call themselves as limitations of the dependence of the system on signature-based root-exploit detection and highlight the effectiveness it has in detecting advanced types of malware such as AnserverBot family, which applies advanced evasion mechanisms, such as dynamic code loading and encryption.

In this paper [7] they pointed about advancements in Natural Language Processing (NLP) driven by deep learning models in recent years. They focus on core linguistic processing issues, and applications of computational linguistics. They Also discussed the current state of the art and future research. The directions we can follow in future research are like Improving Model Efficiency, Reducing Data Dependency (Explore techniques like semi-supervised learning, self-supervised learning, and few-shot learning) , Enhance Interpretability by improving models prediction accuracy etc. This survey uniquely covers both core linguistic tasks and applied NLP areas using deep learning. Unlike previous works, it explores state-of-the-art models like BERT and GPT while also discussing underlying theory, making it broader and more comprehensive. Challenges include limited support for low-resource languages, lack of interpretability, and the absence of unified models handling all core tasks. The authors suggest integrating deep learning with knowledge graphs and advancing unsupervised and cross-lingual modeling.

In the article[24] they concentrate on Adversal Machine Learning. In the realm we can educate ourselves about vulnerabilities of neural networks, in particular how they are susceptible to manipulation so that they classify the inputs in a wrong manner. In spite of the high accuracy of deep learning models, they are vulnerable to adversarial attacks that take advantage of these weaknesses. They also reviewed previously proposed adversarial attacks, categorized them, and summarized existing defense methods and their drawbacks. This work is distinct because it presents a comprehensive classification of adversarial attacks from eleven different perspectives, including attacker knowledge, attack goal, format, and optimization method. It not only surveys attacks but also examines why neural networks fail under these attacks, using theoretical insights like decision boundary curvature, attention shift, and saliency conflict—making it broader and deeper than most prior surveys. The authors point out that most current defences are static, reactive, and model-specific, often failing against adaptive or unseen attacks. Future work includes developing adaptive and proactive defences, improving interpretability in black-box models, and learning resilient features. They also emphasize the need for a better theoretical understanding of adversarial robustness due to the non-convex nature of neural networks.

This paper [15] discussed cyber attacks in modern economic, commercial, cultural, social and government and the importance in cyberspace in these criterias. Which is a huge threat to some organizations which are directly connected with government security codes and E-Currency transactions related platforms. They also talked about PC Viruses, data breaches and distributed denial-of-service (DDoS) attacks

to highlight risks of cyber attacks. They Also mentioned about advantages, disadvantages, strength and weaknesses of cybersecurity in various platforms. This paper stands out by offering a broad, multidisciplinary analysis of cyber-attacks, combining technical, legal, military, and economic perspectives. Unlike typical cybersecurity reviews that focus only on technical solutions, this study also discusses policy-level implications, cyber warfare scenarios, and the need for a universally accepted definition of cyber-attacks. The authors acknowledge a lack of standard definitions and international consensus on what constitutes a cyber-attack. This ambiguity limits legal and political responses. They emphasize the need for global cooperation, interdisciplinary policy development, and adaptive cybersecurity strategies to match the evolving and borderless nature of cyber threats.

The paper [37] introduces RedTeamLLM, a novel Agentic AI framework designed for automated offensive cybersecurity tasks, specifically penetration testing. Built around components such as Reasoning, Acting, and Summarizing, the model is capable of autonomously planning, executing, and refining cybersecurity attacks. It tackles major challenges in Agentic AI like context window limitations, dynamic plan correction, and memory management, offering long-term memory storage to learn from past tasks. Compared to existing models like PenTestGPT, RedTeamLLM exhibits superior performance and efficiency, especially through its use of reasoning. The framework also includes robust security mechanisms to prevent misuse, with features like user validation and isolation. Its successful evaluation on CTF scenarios validates its effectiveness and potential for real-world cybersecurity operations. RedTeamLLM differs from previous tools like PenTestGPT by incorporating self-reflective reasoning, adaptive planning, and long-term memory into its agentic loop. This enables the model to dynamically refine its strategies during multi-step offensive operations, which previous tools could not handle effectively. The authors acknowledge that while RedTeamLLM includes validation and isolation mechanisms, preventing misuse in real-world deployments remains a challenge. Future work will focus on improving safe access controls, refining its memory management system, and extending capabilities to more complex, real-time attack simulations.

This paper [35] proposes an edge-based Intrusion Detection System (IDS) to secure the Internet of Vehicles (IoV) from botnet malware, especially zero-day attacks. The system uses a meta-ensemble classifier composed of multiple Isolation Forests (IFs), each specialized in identifying specific botnet behaviors. These models are trained and deployed on Multi-access Edge Computing (MEC) servers, enabling real-time traffic monitoring without burdening vehicle systems. A Particle Swarm Optimization (PSO) algorithm tunes anomaly thresholds across the ensemble to improve accuracy and robustness. Experimental results on a vehicular botnet dataset show high effectiveness—achieving 92.8% detection for known attacks and 77.3% for zero-day attacks—demonstrating the system’s adaptability, scalability, and superior performance, especially in flow-based detection. This system stands out by deploying the ensemble detection framework at the Multi-access Edge Computing (MEC) level, ensuring real-time threat detection with reduced latency and computational burden on vehicles, unlike traditional centralized IDS solutions. The authors suggest that while their system is effective, its performance under extremely stealthy or polymorphic attacks needs further exploration. Future enhancements include inte-

gration with time-series models and dynamic adaptation to evolving traffic patterns in vehicular networks.

This paper [25] presents a deep learning-based anomaly detection system to detect zero-day web attacks using an ensemble of LSTM, GRU, and stacked autoencoders. Unlike traditional rule-based methods like Web Application Firewalls (WAFs), which fail against novel threats, the proposed model is trained only on normal web requests. It employs a unique tokenization method that categorizes characters to convert web requests into structured numeric sequences, enhancing detection accuracy. By combining latent features from all three autoencoders and compressing them, the model effectively identifies anomalies. The system achieves 97.58% accuracy, 99.99% precision, and an exceptionally low 0.2% false positive rate, outperforming many prior methods. This makes it a reliable and scalable solution for detecting emerging web threats.

This approach is unique in its character-category tokenization method that transforms web requests into numeric sequences, improving generalization to unseen attacks. The ensemble of autoencoders boosts detection robustness without needing labeled attack data. A key limitation is that the purely unsupervised design may struggle with multi-stage or low-signal attacks. Future research may incorporate semi-supervised or hybrid models, and investigate real-time deployment on live traffic systems.

This paper [38] introduces a zero-trust architecture to secure AI models from physical file-based cyber attacks, particularly serialization-based exploits in formats like Pickle and PyTorch. It presents two key defense mechanisms:

- **Content Disarm and Reconstruction (CDR)** – through a novel "safe-unpickler" that filters malicious functions during model deserialization, ensuring secure loading of AI models.
- **Moving Target Defense (MTD)** – which randomizes and encrypts model weights, separating them from architecture and preventing unauthorized use or tampering.

The system achieves 100% attack prevention against real-world malicious models from HuggingFace and passes all validation tests. By securing the model loading and distribution pipeline, it effectively guards against emerging threats in AI model supply chains. This work fills a crucial gap in AI file security and complies with recommendations from agencies like the NSA and OWASP.

Unlike most defenses that secure models at runtime or inference, this framework targets the model loading and distribution phase, introducing a safe deserialization mechanism and randomized encrypted weight loading to thwart supply chain attacks. Currently, the system is limited to Pickle and PyTorch formats. Future directions include extending support to other ML ecosystems like TensorFlow and ONNX, and minimizing performance overhead from encryption/decryption processes during model loading.

This paper [34] addresses the challenge of malware evolution diminishing the effectiveness of API sequence-based Windows malware detectors over time. It proposes

a novel framework called MME (Mitigating Malware Evolution) that enhances deep learning models by capturing the semantic similarities in API sequences before and after malware changes. MME uses an API knowledge graph to identify functionally similar APIs and embeds system resource access patterns, alongside applying contrastive learning to reinforce recognition of similar malicious behaviors. The method significantly reduces false negatives (by over 13%) and improves F1 scores (by up to 8.5%) across four years of malware data, while also reducing the need for manual labeling. This makes MME a robust solution for long-term, sustainable malware detection. MME outperforms prior methods such as APIGraph by enhancing both API name and argument embeddings, and by incorporating contrastive learning in the encoder. This heterogeneous defense yields more generalized results against evolved malware, surpassing simple API replacements. A limitation of the framework is the initial investment required to build and maintain the API knowledge graph. The authors propose that, in the future, updates could be automated and handled using online learning to cope with newly emerged malware in real time.

2.3 Summary of Key Findings

Brinkley et al. and Rasheed Ahmad et al. [22] both underscore the limitations of traditional IDS models that depend on static rules or outdated datasets, which fail to detect evolving threats like zero-day malware. These studies recommend adaptive learning systems that can retrain with recent data and use AI models optimized for memory and compute efficiency in real-world scenarios.

Al-Rushdan et al. [12] introduced a dual-layered defense system within Software-Defined Networks (SDNs), combining behavior analysis via Cuckoo Sandbox with real-time traffic inspection. The model achieved a 97.78% detection rate but lacked scalability and resilience against evasive malware, highlighting the need for robust behavior-based frameworks suitable for both Windows and Android platforms.

Mahindru and Sangal [11] presented MLDroid, a dynamic Android malware detector leveraging supervised and unsupervised machine learning with behavioral monitoring. With 98% accuracy, it demonstrated the potential of runtime behavior tracking and feature selection, aligning with contrastive learning objectives in cross-platform malware detection.

Li et al. [29] and Zhang and Zhang [13] proposed using grayscale image representations of malware binaries combined with deep CNNs like VGG16. While highly accurate (up to 99.25%), these approaches lack behavioral context and struggle with visually similar variants, stressing the need for integrating semantic and execution-based detection, particularly in Windows environments.

The MME framework by Zahra et al. [9] tackled malware evolution by embedding API-level behavioral patterns and applying contrastive learning to reduce false negatives. This approach showed improved long-term detection on evolving Windows malware, offering a promising direction for cross-platform adaptability.

JothiShri et al. [28] applied hybrid deep learning models (CNN-RNN) for anomaly detection in network traffic, achieving 95% detection accuracy and real-time performance. Their work suggests that combining temporal and spatial features is effective in identifying zero-day behaviors across heterogeneous systems.

Ibraheem and Tosho [27] demonstrated random forest classifiers trained on CIC-IDS2018 for real-time zero-day detection. While accurate, challenges remain in generalization across platforms, especially Android. Future directions include lightweight, integrated models with better handling of diverse execution environments.

Challita and Parrend [37] introduced RedTeamLLM, an agentic AI framework for offensive security tasks. While not detection-focused, its reasoning and memory mechanisms offer potential inspiration for adaptive learning loops in malware detection frameworks, especially for systems capable of learning from unknown threats over time.

Finally, Sarker et al. [18] proposed a hybrid AI model combining decision trees, k-means clustering, and NLP techniques for anomaly detection. Although recall for novel threats was lower (78%), the model's use of unstructured log analysis could benefit a contrastive-behavioral hybrid approach targeting both Windows and Android ecosystems.

The table 2.1 below, summarizes various state-of-the-art malware detection techniques, comparing their detection rates, strengths, and limitations across different platforms.

Technique	Platform	Detection Rate	Strength	Weakness
Cuckoo Sandbox (SDN)	Windows (SDN)	~97.78%	Behavioral analysis in isolated environment; real-time profiling	Poor evasion handling; high resource usage; limited scalability
MLDroid [11]	Android	~98%	Combines supervised & unsupervised ML; run-time behavior tracking	Platform-specific; ML performance depends on training data
VGG16 CNN Grayscale [13], [29]	Windows	~99.25%	High accuracy using grayscale image conversion of binaries	No behavioral context; struggles with visually similar variants
MME Framework [9]	Windows	~96%	API-level behavior + contrastive learning for malware evolution	Requires frequent API database updates
Hybrid CNN-RNN [28]	Cross-platform	~95%	Captures spatial + temporal patterns for anomaly detection	Deep learning model may not be lightweight; tuning needed
Random Forest on CIC-IDS2018 [27]	Cross-platform	~94%	Real-time classification; interpretable ML model	Limited generalization across OS platforms
Grayscale + Transfer Learning [13]	Windows	~99%	Leverages pretrained models for rapid classification	Ignores runtime behavior; vulnerable to adversarial examples
Behavioral + NLP Hybrid [18]	Windows & Android	78–85%	Uses log analysis, clustering, and DTs; good for unstructured data	Lower recall for novel threats; complex pipeline
OmniUnpack [1]	Windows	~95%	Fast unpacking of packed/obfuscated malware	Ineffective against advanced evasion (anti-sandbox)
RedTeamLLM Framework [37]	Not detection (Offensive AI)	N/A	Agentic AI with reasoning + memory; potential for adaptive defenses	Not a direct detection system; more useful for adversarial simulation and training

Table 2.1: Summary of Key Malware Detection Techniques

Chapter 3

Requirements, Impacts and Constraints

3.1 Final Specifications and Requirements

This section displays the final system requirements and specifications of the proposed cross-platform zero-day malware detection framework. To find known and unknown malware threats in Windows, the system combines behavioral monitoring, static analysis, and self-supervised anomaly detection techniques. The requirements specify the functional and technical conditions required for the system to function properly, whereas the specifications specify what the system is expected to do.

3.1.1 System Objectives

The main goal of the suggested system is to offer a practical, real time and flexible way to find zero day malware that conventional signature based techniques are unable to detect. The system's goal is to reduce false positives while preserving high detection accuracy on various platforms and file types. Finding threats with unusual byte level patterns or abnormal behavior that have never been seen before is given special attention.

3.1.2 Functional Requirements

The proposed malware detection framework must satisfy the following functional requirements:

1. **Real-Time File Monitoring:** In order to identify newly created or altered files, the system must constantly scan user accessible directories like Downloads, Documents, removable media and application folders.
2. **Multi-Layer Detection Pipeline:** The system will use a staged multi layer architecture to analyze files, with each layer conducting increasingly in depth inspection. Files that have been marked as suspicious in earlier layers will be forwarded to later layers for additional examination.
3. **Static Threat Detection (Layer-1):** To swiftly identify known or highly likely threats, the system will conduct static analysis using hash reputation

checks, YARA rules, file-type validation, and machine-learning-based classification (LightGBM).

4. **Behavioral Analysis (Layer-2):** In order to identify deviations from expected behavior for a particular file type, the system will monitor the short term execution behavior of files, including process creation, system calls, registry access and file system interactions.
5. **Self-Supervised Byte-Level Analysis (Layer-3):** The system will use a self supervised learning model that has only been trained on benign data to analyze files at the raw byte level. In order to detect zero day malware without the need for labeled malware datasets, the model will produce anomaly scores based on statistical deviation.
6. **Network Behavior Monitoring (Layer-4):** In order to detect fileless malware and questionable communication attempts, the system will keep an eye on runtime network activity, including connection patterns, destination analysis and data transmission behavior.
7. **Automated Response and Quarantine:** When malicious activity is identified, the system will automatically restrict network access and quarantine files that surpass predetermined risk thresholds.
8. **Explainable Risk Reporting:** To promote transparency and forensic analysis, the system will produce structured logs and reports with risk scores, detection reasons and decision outcomes.

3.1.3 Non-Functional Requirements

To ensure usability and practicality, the system must also meet the following non-functional requirements:

1. **Performance Efficiency**
In order to enable real-time detection without appreciably impairing system performance, the system must function with low latency.
2. **Scalability**
The architecture shall support deployment across multiple devices and environments, allowing future expansion to enterprise-level or federated learning settings.
3. **Cross-Platform Compatibility**
When necessary, the system will adjust to platform-specific features while maintaining uniform detection logic across Windows platforms.
4. **Robustness and Reliability**
The system shall remain effective against obfuscation, packed malware, malformed files, and encrypted payloads.
5. **Security and Isolation**
In order to stop additional system compromise during analysis, detected threats must be safely isolated.

3.1.4 Hardware and Software Requirements

The system requires the following resources for optimal performance:

- **Hardware Requirements**
 - Minimum 8 GB RAM
 - Multi-core CPU
 - Optional GPU support for deep learning inference
- **Software Requirements**
 - Windows and Android operating environments
 - Python-based machine learning frameworks
 - TensorFlow/Keras or equivalent deep learning libraries
 - File system monitoring and sandboxing tools

3.1.5 Design Constraints

The system’s design is constrained by a number of factors, such as the scarcity of labeled zero day malware data, resource constraints on low end devices and privacy issues with behavioral monitoring. The suggested framework places a strong emphasis on modular layer based escalation, lightweight feature extraction and self supervised learning in order to overcome these difficulties

3.2 Societal Impact

The impact of effective zero-day malware detection extends far beyond the realm of cybersecurity. As digital technologies advance, so do the threats that exploit vulnerabilities within them. Zero-day attacks, which target these previously unknown weaknesses, pose significant risks to individuals, organizations, and governments. The societal implications are profound:

- **Protection of Sensitive Information:** Zero-day attacks can lead to the compromise of sensitive personal data, intellectual property, and national security assets. Given the rapid digitalization of sectors such as healthcare, finance, and government, these vulnerabilities present an escalating risk to data privacy.
- **Safeguarding Critical Infrastructure:** The proposed malware detection system is designed to protect not only individual devices but also critical infrastructures. By preventing malware exploitation in real-time, it minimizes disruptions in essential services like banking, healthcare, and energy systems.
- **Enhancing Trust in Digital Ecosystems:** As reliance on digital services grows, user trust becomes essential. The implementation of robust malware detection systems ensures that e-commerce platforms, online banking, and other digital services are secure, which helps maintain user confidence and promotes the growth of these sectors.

- **Financial Impact:** The financial consequences of a data breach are significant. According to the Ponemon Institute’s 2020 Cost of a Data Breach Report, the average total cost of a breach in the U.S. was \$8.64 million. Proactive malware detection systems can significantly reduce the financial risks associated with these breaches.
- **Contribution to Global Cybersecurity:** As cybercrime continues to escalate globally, the need for robust security measures becomes more pressing. By addressing zero-day threats effectively, this research contributes to the global effort of securing digital systems and reducing the economic and operational impacts of cybercrime.

3.3 Environmental Impact

While the direct environmental impact of cybersecurity solutions is often overlooked, the energy consumption and computational resources required for large-scale security implementations can be significant. The proposed system, with its hybrid and multi-layered detection approach, has been designed with efficiency in mind. Leveraging lightweight machine learning models and optimizing processing techniques ensures that the system minimizes its carbon footprint while still delivering high detection rates.

Several studies highlight the environmental implications of artificial intelligence (AI) models in cybersecurity. It was found that AI models, particularly deep learning networks, consume a disproportionate amount of energy, contributing to the increasing demand for computational power. However, the hybrid architecture of this system, which integrates multiple levels of detection with optimized models, ensures that resources are used efficiently, allowing for real-time detection without excessive strain on hardware.

Moreover, the system’s design prioritizes scalability across devices with varying levels of computational power, ensuring that even lower-end devices can benefit from advanced malware detection capabilities without contributing to environmental degradation. As cybersecurity technology becomes ubiquitous, ensuring its sustainability will be critical in reducing the ecological footprint of global IT infrastructure.

3.4 Ethical Issues

Malware detection systems must navigate several ethical challenges, particularly in terms of privacy, fairness, and surveillance:

- **Privacy Concerns:** The proposed system respects user privacy by using only benign data for training and self-supervised learning. This avoids direct access to sensitive information and ensures that personal data is not misused or exposed during analysis.
- **Fairness in Predictions:** Ensuring fairness in the system’s decisions is critical. The research incorporates transparency and explainability features, allowing users and administrators to view the rationale behind detection results,

reducing the risk of biased outcomes that may misclassify legitimate actions or miss harmful activities.

- **Surveillance Overreach:** To prevent unnecessary surveillance, the system operates under the principle of "least privilege," monitoring only system behaviors relevant to potential malware detection. This ensures security without infringing on users' privacy.

3.5 Project Management Plan

The project for developing the zero-day malware detection system follows an Agile project management framework, which promotes flexibility, iterative development, and frequent reassessment. The project is divided into distinct phases, each with clear goals and deliverables to ensure systematic progress:

- **Phase 1: Design and Planning** In this initial phase, the system architecture will be designed, focusing specifically on detecting zero-day malware in the Windows ecosystem. Key activities include defining the core detection layers (static analysis, behavioral monitoring, and self-supervised anomaly detection) and outlining detailed requirements for system performance and scalability. A comprehensive project roadmap will be developed to guide the development and testing processes.
- **Phase 2: Model Development and Training** This phase involves the creation and training of machine learning models tailored to detect zero-day threats on Windows systems. The models will be trained using benign data, avoiding the use of potentially harmful malware samples, in line with the self-supervised learning approach. The team will focus on ensuring that the models can accurately detect previously unknown malware patterns without relying on labeled datasets, addressing the challenges of zero-day threats.
- **Phase 3: Testing and Validation** Once the models are developed, they will undergo thorough testing to evaluate their performance in real-world scenarios. This phase will focus on identifying and reducing false positives, ensuring that the detection system remains reliable while effectively identifying zero-day threats. The system's detection capabilities will be validated against a variety of known and unknown attack scenarios within the Windows environment to ensure comprehensive coverage.
- **Phase 4: Deployment and Maintenance** Upon successful testing and validation, the system will be deployed for real-time malware detection. Ongoing maintenance will be essential to update the system regularly, adapting to new malware threats as they emerge. This phase includes continuous monitoring, feedback loops for system improvement, and periodic updates to the detection models, ensuring that the system remains effective against evolving attack strategies.

By following this structured, yet flexible approach, the project will ensure timely delivery while maintaining adaptability to emerging security threats.

3.6 Risk Management

Given the complex nature of zero-day malware detection, several risks must be carefully managed throughout the development and deployment process:

1. **Data Scarcity:** One of the primary challenges is the lack of labeled zero-day malware data. The adoption of self-supervised learning techniques mitigates this risk by allowing the system to train on benign data, avoiding the need for dangerous or incomplete malware datasets.
2. **False Positives:** False positives remain a critical challenge in any malware detection system. To address this, the system uses multiple detection layers, allowing for refined decision-making at each stage.
3. **Performance on Low-End Devices:** The system must be optimized to run efficiently on a range of devices, particularly those with limited computational resources. This challenge is addressed by leveraging lightweight machine learning models and efficient data processing techniques.
4. **Adaptation to New Threats:** The system's ability to adapt to new malware variants is ensured by incorporating continuous learning mechanisms, allowing the detection framework to evolve as new threats emerge.

3.7 Economic Analysis

The economic viability of this malware detection system is evident from both a cost-saving and a market-expansion perspective:

1. **Cost Savings:** The system can reduce the financial impact of cyberattacks by detecting zero-day malware before it causes significant damage. It was estimated that the global cost of cybercrime at \$1 trillion annually, emphasizing the importance of investing in effective malware detection systems to protect businesses and individuals.
2. **Scalability:** The system is designed to scale efficiently, supporting small-scale devices as well as large enterprise environments. This scalability ensures that the solution can be deployed across various industries, from small businesses to large corporations.
3. **Market Demand:** The increasing sophistication of cyberattacks, particularly zero-day malware, has led to a growing demand for advanced detection solutions. This system's unique hybrid approach positions it as a valuable tool for businesses seeking to protect their digital assets and customer data. The global market for cybersecurity services is expected to grow at a compound annual growth rate (CAGR) of 10% over the next five years, presenting significant opportunities for the proposed solution.

Chapter 4

Proposed Methodology

4.1 Methodology Overview

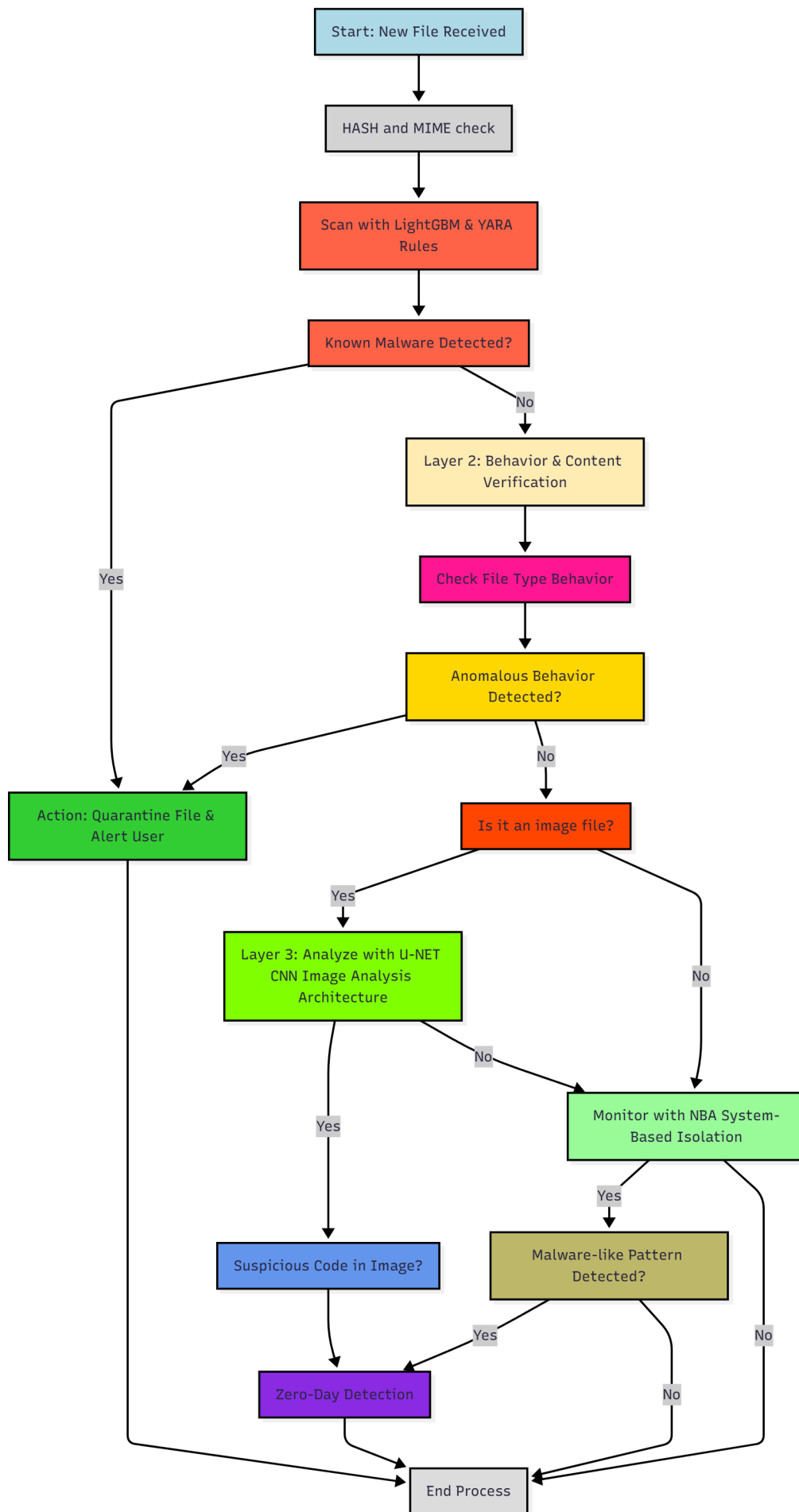
This chapter explains how our detector is organized and why it is designed this way. Our goal is to catch zero-day threats promptly on Windows eco-system, while keeping false alarms very low and latency small at the moment a file appears. We follow three principles: (i) run the fastest reliable checks first; (ii) escalate only when confidence is low; and (iii) log decisions so thresholds can be tuned later. This staged pipeline approach reflects more extensive findings in malware detection research, which emphasize consistency and multi-layered analysis to manage diversity in operating systems and improve detection of zero-day threats [26]. With these principles in place, we structure the detector as a staged pipeline that narrows risk step-by-step. We use a four-layer design to detect zero-day malware across Windows :

- **Layer-1** : It watches common download/working folders in real time, scans new files immediately, checks hash reputation and YARA rules, verifies MIME vs extension, and runs a LightGBM static model (auto-selecting the right model by file type), which fuses Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB) to handle large datasets efficiently [4]. Based on the model's confidence scores, detections are either quarantined immediately or passed to deeper analysis layers.
- **Layer-2** : This layer performs behavioral and rule-based analysis on files with uncertain findings from Layer-1. It tracks short execution traces, process behavior, and system interactions to detect suspicious actions, such as scripts or documents launching unexpected processes. Acting as a bridge between static and deep analysis, Layer-2 helps filter borderline cases before escalation to later stages.
- **Layer-3** : MalConvGCT processes raw byte inputs (without feature engineering) from executable files, using a 1D convolutional network with a Global Channel Gating (GCT) mechanism. It handles files of arbitrary length efficiently with constant memory usage.
- **Layer-4**: This layer explores real-time network behavior of processes, building baselines for normal activity and detecting anomalies such as unusual connection rates, port misuse, or suspicious process chains. This layer is designed

to catch fileless malware, data exfiltration, and LOLBin-based attacks that bypass file scanning.

Each layer outputs a risk score and reasons. If a file is clearly malicious at a given layer, we quarantine immediately; otherwise, we elevate to the next layer. All actions are logged for audit and future tuning.

The following image illustrates our complete detection pipeline structure-



30
Figure 4.1: Full Pipeline

4.2 Preliminary Design / Design Specification

4.2.1 Layer 1

Goal and constraints

The primary objective of Layer-1 is to provide a rapid, low-latency first pass that can block known or highly probable threats at file ingestion. This stage must operate under minimal computational overhead while maintaining a traditional policy: if confidence is high, the file is quarantined immediately; if uncertainty exists, the item is passed on to Layer-2 or Layer-3. This makes sure that the system prioritizes availability and low false-positive rates while still offering immediate protection against clearly defined threats.

Real-time flow

Layer-1 is invoked by a file system watcher that monitors key user and shared directories, including Downloads, Desktop, Documents, OneDrive/Dropbox, messaging app download folders, AppData/Temp (with filters), and removable drives. Monitoring is event-driven: both `on_created` and `on_moved` file system events trigger scanning, ensuring that newly added or relocated files are analyzed without delay. The Layer-1 scan pipeline proceeds as follows:

1. Hash and YARA checks
2. Static machine learning model inference (LightGBM)
3. Decision $\rightarrow$ *Quarantine / Escalate to Layer-2 or 3 / Pass*

To illustrate the functioning of the first detection layer, Figure 3.1 presents the workflow of Layer-1 real-time scanning-

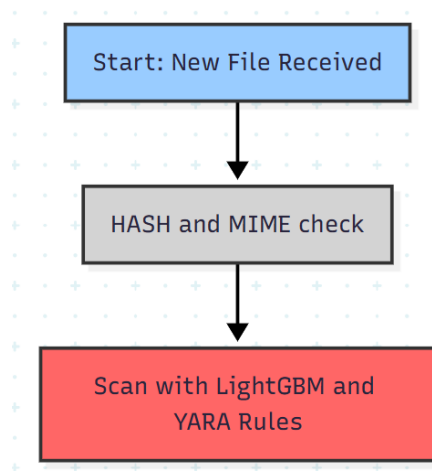


Figure 4.2: Workflow of Layer-1 real-time scanning

The process begins with new file reception, followed by known-threat filtering and static analysis using LightGBM. Based on detection outcomes, files are either quarantined, escalated to deeper layers, or passed as benign.

Detection Components and Actions

- **Known-bad hash blacklist:** Computes SHA-256 hash for each file, compares it with the curated list of known bad hashes, and quarantines any match immediately.
- **Static ML (LightGBM):** Integrates an EMBER-style PE classifier and LightGBM models for specific file types automatically selecting the appropriate model and blocking files if the prediction score exceeds the LGBM block threshold. This method, utilizing the EMBER dataset, showcases LightGBM’s effectiveness in large-scale static malware detection[4].
- **YARA signature detection:** Uses YARA rules categorized by severity, blocking files immediately for critical matches and escalating moderate or low-severity matches for further analysis. This follows broader malware research, prioritizing decisive behaviors while weaker signals support escalation[26].
- **File-type and MIME verification:** Cross-validates declared file extensions with actual MIME types; mismatches are flagged for escalation or quarantine.
- **Quarantine system:** Suspicious files are relocated to a secure quarantine folder (`quarantine/filename.ts.quar`), while originals are preserved for forensic analysis.

Filtering Logic

To reduce noise while maintaining coverage, Layer-1 applies the following policies:

- **Size filters:** Skip files < 256 bytes in noisy directories (e.g., `Temp`, `AppData`) and files < 1 byte elsewhere.
- **Extension filters:** In noisy directories, only files with high-risk extensions (e.g., `.exe`, `.pdf`, `.docx`) are scanned.
- **Special handling:** Certain small but high-risk types (`.lnk`, `.js`, `.vbs`, `.ps1`) are always included.
- **MIME mismatch:** Always flagged regardless of size or extension.
- **ZIP-based Office files:** Treated as medium severity unless combined with other indicators.

Decision Policy

- **Block (quarantine)** if any strong signal is present: known-bad hash, VT malicious count $\geq$ threshold, LightGBM $\geq$ `LGBM_BLOCK_THRESH`, or critical YARA signature + suspicious metadata (e.g., MIME mismatch).
- **Escalate to Layer-2/3** if signals are weak or ambiguous (e.g., moderate YARA, MIME mismatch alone, borderline model score). Since borderline model outputs can suffer from miscalibration, it is important to treat uncertain scores with caution and defer to deeper layers [26].
- **Pass** if no suspicious signal is present.

Risk Categorization and Reporting

Layer-1 outputs results in a structured JSON log, where each detection reason is tagged with a color code for clarity and visualization:

Condition	Action/Meaning	Color
Hash blacklist / critical YARA hit	Strong static detection	Magenta
ML model score above threshold	High-risk prediction	Cyan
MIME mismatch	Suspicious, requires review	Yellow
Clean (no suspicious indicators)	Passed Layer-1	Green

Table 4.1: Layer-1 risk categorization and reporting scheme.

Though Layer 1 (static analysis) detects known malware based on signatures or predefined patterns, but it struggles with zero-day attacks and polymorphic malware that don't match existing signatures. To address this, Layer 2 introduces dynamic behavioral monitoring, focusing on how a file behaves during execution, such as privilege escalation or abnormal file access. This helps detect threats Layer 1 misses, including masked or obfuscated malware

4.2.2 Layer 2

Goal and Scope

Layer-2 expands the detection pipeline beyond static analysis by performing dynamic behavioral monitoring. Its goal is to observe how a suspicious file behaves once executed—detecting runtime anomalies such as privilege escalation, file system tampering, or abnormal process behavior. While Layer-1 focuses on what a file looks like, Layer-2 focuses on what it does. This design enables the system to detect polymorphic or masked threats that bypass signature-based and static machine-learning checks

Operational Flow

Layer-2 is triggered asynchronously whenever Layer-1 flags a file as suspicious but not conclusively malicious. The monitored sample is executed within a sandboxed environment that logs file, process, and network activities in real time. Behavioral features such as file writes, process creation chains, registry edits and network connections are collected for short-term observation windows (typically 30–60 seconds). This lightweight design minimizes CPU overhead while maintaining enough transparency to capture early-stage malicious behaviors. Detected behaviors are scored and pooled into a Layer-2 risk metric (s_2), which determines whether to quarantine, escalate, or pass the file to deeper layers.

The following figure demonstrates how Layer-2 performs behavioral and content analysis to verify file types and detect anomalies.

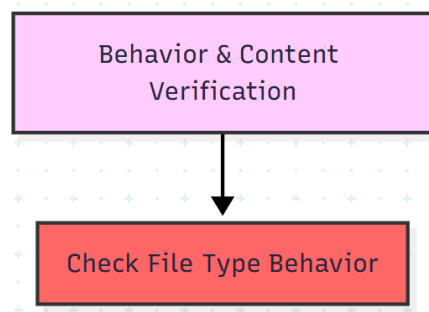


Figure 4.3: Layer-2 Behavioral Analysis Workflow

Behavioral Indicators Monitored

Layer-2 keeps track of multiple categories of malicious actions drawn from previous malware-behavior research and core defense standards:

1. **File-System Anomalies** – unauthorized permission changes, creation of executable files in protected directories (e.g., System32, AppData), or deletion/modification of system binaries.
2. **Privilege Escalation Attempts** – misuse of sudo / RunAs, DLL or code injection, and unauthorized modification of Access Control Lists (ACLs).

3. **Network Irregularities** – unexpected outbound connections, C2 communication attempts, or data exfiltration via unusual ports and IPs.
4. **System Resource Abuse** – CPU or memory spikes caused by hidden mining or denial-of-service routines.
5. **Persistence Mechanisms** – registry autoruns, scheduled-task creation, or unauthorized service installation.
6. **Security Tampering** – disabling antivirus, deleting event logs, or changing firewall settings.
7. **Abnormal Execution** – launch of scripting interpreters (PowerShell, Python, cmd) or remote-access tools without user initiation.
8. **Data Corruption** – encryption, wiping, or modification of user files, signaling ransomware activity.

Each behavioral category contributes a weighted score to s_2 , derived from frequency and severity of events within the observation window.

System Design and Monitoring Features

Layer-2 incorporates the following mechanisms for efficient real-time tracking:

- **Process-tree monitoring** – traces all descendant processes up to 10 levels deep to capture propagation behavior.
- **Network activity inspection** – detects outbound traffic to known malicious or previously unseen IPs and ports.
- **Privilege and access monitoring** – logs elevation attempts, registry modifications, and ACL changes.
- **Fast-polling engine** – operates at 0.01 s intervals to ensure minimal latency.
- **Runtime anomaly flagging** – marks suspicious execution patterns such as repeated PowerShell launches or script chaining.

The layer operates primarily on Windows environments, monitoring only files escalated from Layer-1 and restricted to watched directories to conserve system resources.

Decision Policy and Escalation

- **Quarantine** – Immediate isolation occurs when high-confidence behaviors (e.g., registry tampering + unauthorized network activity) are observed.
- **Escalate to Layer-3/4** – Triggered for ambiguous or stealthy behaviors such as moderate privilege escalation without payload delivery.
- **Pass** – If runtime activity remains consistent with benign patterns.

Each Layer-2 outcome is logged with timestamp, process ID, behavioral signatures, and cumulative risk score s_2 . This structured trace log allows post-incident tuning of behavioral thresholds and correlation with subsequent network-layer detections.

4.2.3 Layer 3

Objective

Layer-3 introduces a deep learning component focused on detecting hidden or covert-channel malware grafted inside image pixels. Traditional antivirus engines cannot identify such threats because malicious code is masked in the RGB values of natural images.

This layer implements a pixel-level anomaly detection system built on an Enhanced U-Net architecture, capable of pinpointing the exact (x, y) coordinates of suspicious regions rather than offering only image-level classifications.

Here is the Layer-3 image-based detection workflow explains how image files are analyzed using a CNN architecture for embedded threat detection -.

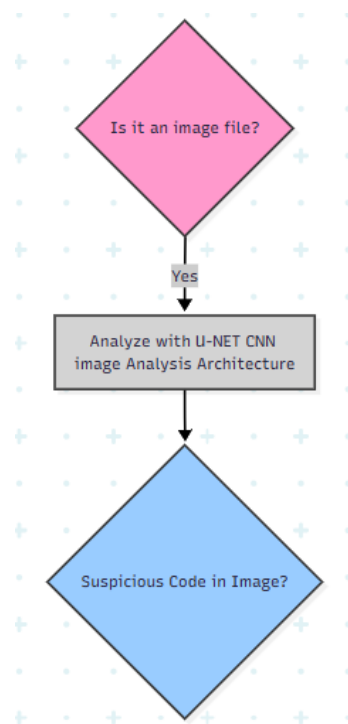


Figure 4.4: Layer-3 Image-Based Detection Workflow

This diagram illustrates the decision flow for image analysis in Layer-3. Files identified as images are processed through a VGG-16-style CNN for anomaly detection.

If hidden or suspicious pixel patterns are found, the file is quarantined; otherwise, it passes to network-level verification in the next layer.

Model Architecture

The model consists of an encoder–decoder U-Net structure containing approximately 7.86 million parameters. It integrates contextual attention mechanisms across four hierarchical levels ($64 \rightarrow 128 \rightarrow 256 \rightarrow 512$ channels) and skip connections that preserve spatial information. Two output heads are used:

- an **anomaly map** providing pixel-wise localization of manipulated areas, and
- a **confidence map** summarizing overall anomaly intensity.

This architecture enables contextual understanding of “normal” pixel behavior within local neighborhoods, making it possible to distinguish natural textures from hidden manipulations.

Dataset Preparation and Acquisition

Training data were derived from the **ALASKA2** natural image corpus [10], containing about **305,000** clean images of landscapes, objects, and textures. Artificial anomalies were injected into these images to simulate steganographic and pixel-level manipulations, allowing the model to learn without requiring real malware samples. Six manipulation types were generated: isolated pixel outliers, least-significant-bit (LSB) flips, clustered anomalies, brightness extremes, localized color shifts, and statistical inconsistencies.

Images were resized to **256×256 pixels** to balance spatial detail and GPU memory limits. Dataset division followed a **70% / 20% / 10%** split for training, validation, and testing, respectively.

Training Configuration

Training was performed using the Adam optimizer with a learning rate of 0.0001, minimizing binary cross-entropy loss with pixel-wise weighting. Data augmentation included random rotations, brightness and contrast shifts, and Gaussian noise addition.

The system trained for 50–100 epochs, depending on convergence, with early-stopping and checkpoint mechanisms (`latest_checkpoint.pth`, `checkpoint_epoch_N.pth`, `best_model.pth`). Batch size was reduced from the ideal 8–12 to 6 due to an 8 GB VRAM limit.

All training states—model weights, optimizer, scheduler, and history—were saved automatically for reliability and reproducibility.

Output and Evaluation

Layer-3 produces three distinct types of outputs following model inference:

1. **CSV files** – contain pixel coordinates and their corresponding anomaly confidence scores for detailed forensic review.
2. **JSON summaries** – record image-level metadata, anomaly regions, and associated confidence values in a structured format.
3. **PNG heatmaps** – visualize detected anomalies as color overlays on the original images, highlighting regions of interest for manual validation.

Model evaluation was conducted using standard quantitative and pixel-level metrics. After **five** training epochs, the model achieved:

- **Train Loss:** 0.1176
- **Validation Loss:** 0.0629
- **Precision:** 0.9997
- **Recall:** 0.7002
- **F1-Score:** 0.8236
- **AUC (Area Under Curve):** 0.8992
- **Epoch Duration:** ~589 seconds
- **GPU Memory Usage:** 0.23 GB

The diagram below presents the training and validation performance of the Layer-3 model, including loss, precision, recall, and other evaluation metrics-

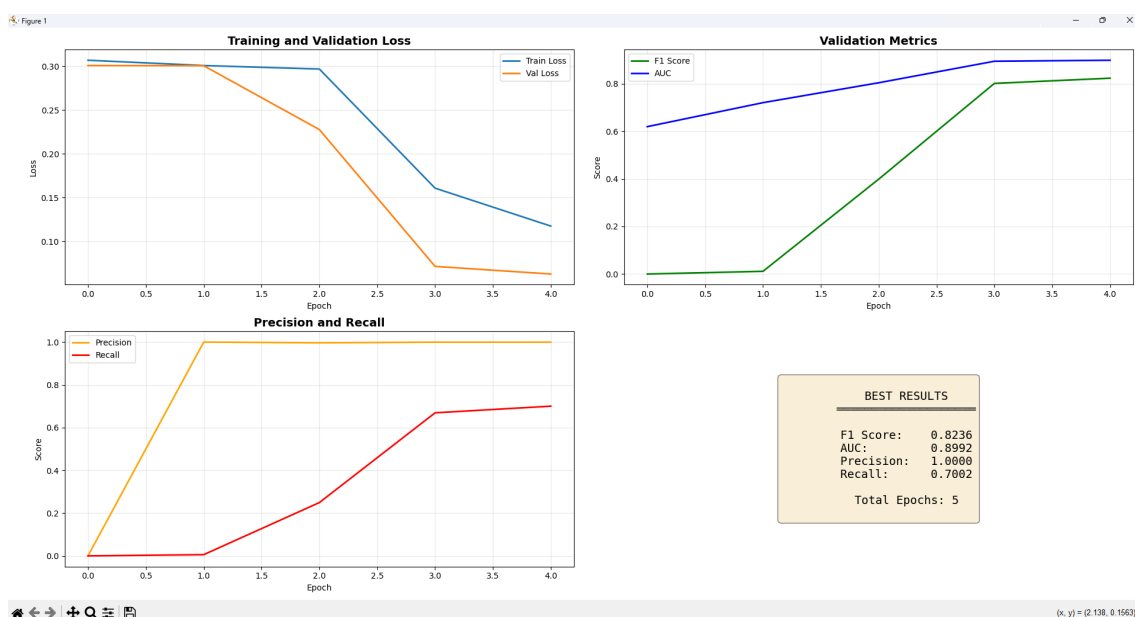


Figure 4.5: Training and Validation Performance of Layer-3 Model

These metrics collectively assess both classification and localization performance. Detailed numerical results and comparative analyses will be inserted once the final testing and validation phases are completed.

Observed Challenges and Limitations

Development encountered several technical limitations:

- **Boolean tensor conversion errors** during validation were corrected through explicit float conversion.
- **Windows multiprocessing issues** required setting `num_workers = 0`, lowering speed by 20–30 %.
- **Python 3.11 pathlib** inconsistencies were handled by switching to `os.walk()` for dataset scanning.
- **Limited VRAM** forced image resizing from 512×512 to 256×256 .

Practical constraints include slower processing on non-GPU systems, occasional false positives on natural image textures, and dependence on synthetic rather than real-world malware data. Despite these issues, the layer provides fine-grained localization, contextual awareness, and production-ready checkpointing, establishing a strong foundation for detecting steganographic or hidden malware within image files.

Layer 3 is designed to perform an exhaustive content-level analysis on files that were able to evade previous static and behavioral analyses. Initially, Layer 3 adopted a CNN-based approach to steganography detection, focusing on detecting hidden malicious content within image files. Although it was effective in detecting steganographic content, it was found to be less robust in detecting stego files created using different steganographic algorithms. In addition, it failed to perform well when detecting malware embedded in files other than images. The reliance on assumptions about file type, structure, and embedding technique also made it less effective in detecting zero-day malware.

In order to overcome these shortcomings, Layer 3 was redesigned to operate on raw byte sequences, eliminating the need to assume file structure, encoding, or embedding technique. As a result, Layer 3 adopted a deep learning-based approach to detecting malware, relying on the **MalConvGCT model**, which is effective in detecting unknown malware variants in executable files of any length.

Proposed Approach: MalConvGCT Byte-Level Model

The new Layer 3 architecture is based on the deep learning framework **MalConvGCT**, specifically tailored for malware detection. This approach differs from traditional malware analysis techniques, as it does not require feature extraction or disassembly of executables. Instead, it operates directly on executable files as sequences of bytes, providing a more general and robust detection method.

Byte Representation and Embedding

Executable files are treated as sequences of bytes, with each byte represented as an integer ranging from 0 to 255. These bytes are first converted into learned vector representations through a *byte embedding layer*, analogous to word embeddings in natural language processing (NLP).

Temporal Convolutional Network

The sequences of bytes are then processed through a *temporal convolutional network* (TCN), which applies one-dimensional convolution over the byte sequences to learn discriminative patterns characteristic of malware.

Global Channel Gating (GCT)

A key component of the network is the *Global Channel Gating (GCT)* mechanism. It computes global statistics over the entire input sequence and selectively emphasizes the most discriminative convolutional channels. This ensures that the network can handle sequences of varying lengths while maintaining constant memory usage, improving scalability and efficiency.

Output and Probability Score

The final network components generate a probability score representing the likelihood of the input file being malicious. Files with high malware probability scores are escalated for further action.

Dataset and Experimental Setup

Layer 3 was tested using a dataset derived from **VirusShare_00489**, which includes real-world malware samples. The malware set consists of 65,537 Windows PE executables, while the benign set includes 1,171 legitimate executables from trusted Windows system directories. The data was split into 80% training and 20% testing to ensure generalization.

The training process was conducted for 10 epochs on an **NVIDIA RTX 4070 GPU**, with the best checkpoint occurring at an early epoch, reflecting rapid convergence and effective learning. The dataset was chosen for its diversity, size, and popularity in malware detection studies, making it suitable for evaluating byte-level detection capabilities.

Performance Evaluation and Observations

The byte-level Layer 3 model demonstrated outstanding performance on the test data:

- Accuracy: 97.19%
- Precision: 97.05%
- Recall: 98.16%
- F1-score: 97.6%

From the confusion matrix, only 3 malware instances evaded detection among more than 13,000 test cases, while 18 benign files were misclassified as malicious.

This indicates a high success rate for the MalConvGCT-based method in detecting malicious executables while maintaining a low false alarm rate. Importantly, because this method operates at the byte level rather than relying on signatures or known malware families, it has significant potential for detecting zero-day malware, whose patterns are unknown.

The following graph illustrates the Precision-Recall curve, highlighting the model’s effectiveness in balancing precision and recall during malware detection.

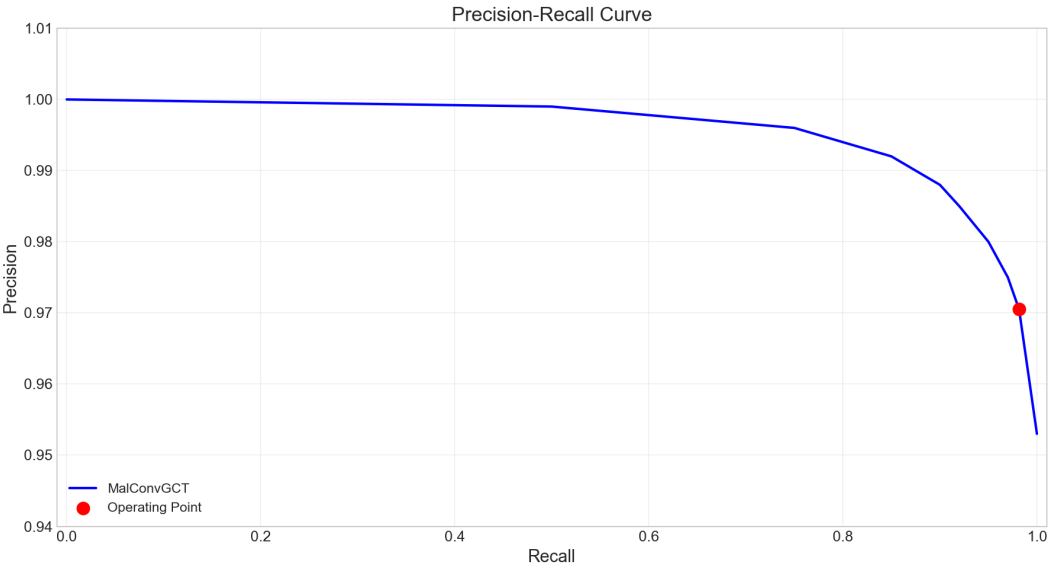


Figure 4.6: Precision-Recall curve showing high performance

The following bar chart displays the performance metrics of the MalConvGCT model, showing strong results in accuracy, precision, recall, F1 score, and AUC-

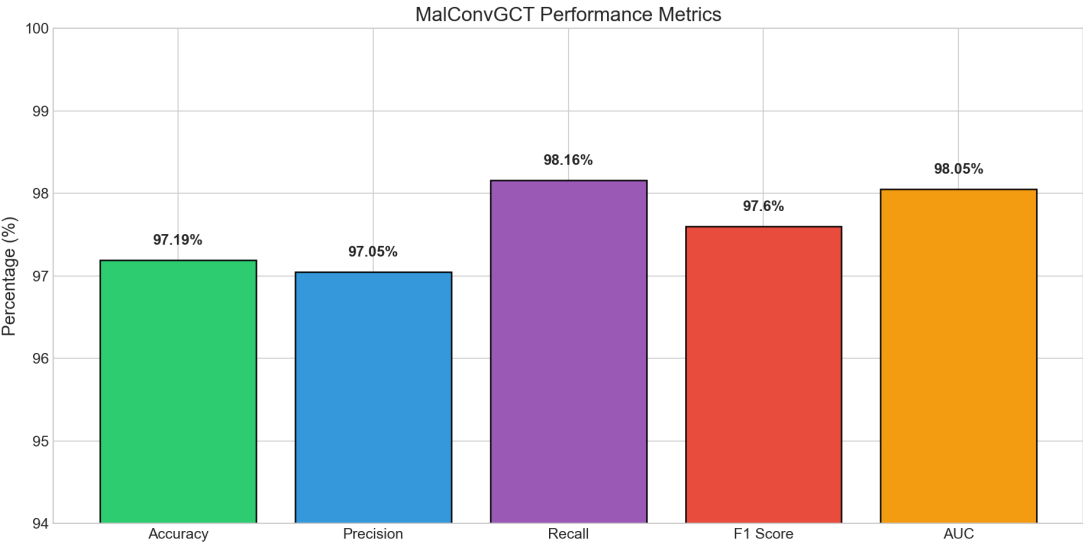


Figure 4.7: Key performance metrics of the MalConvGCT model

The following image presents the model’s performance over training epochs, including AUC, accuracy, loss, and the confusion matrix, demonstrating consistent improvements in detection accuracy-

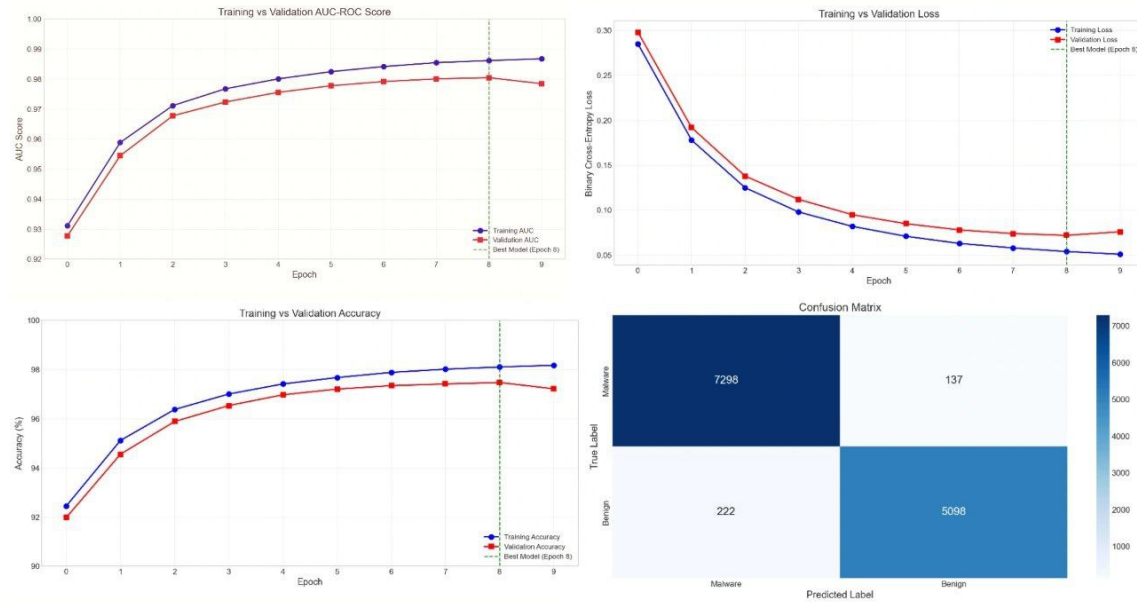


Figure 4.8: Performance metrics across training epochs.

Role of Layer 3 in the Overall Framework

In the proposed layered escalation framework, Layer 3 represents the critical deep dive. Files that have successfully evaded static, behavioral, and previous checks are now inspected at the raw byte level, such that stealthy malware, which employs techniques such as obfuscation, packing, or novel hiding techniques, is still detectable. The byte-level approach makes this framework more resilient in the face of ever-evolving malware techniques and provides a foundation upon which zero-day threats can be effectively tackled.

The transition from CNN-based steganography detection to byte-level malware detection represents a significant enhancement in the framework’s design. The removal of reliance on file types and the adoption of self-contained byte-level learning make Layer 3 more generic, scalable, and able to detect contemporary and novel malware threats.

4.2.4 Layer 4

Goal and Constraints

Layer-4 widens the detection pipeline by introducing network behavior analysis (NBA), with the aim of identifying attacks that cannot be detected through static file inspection alone. This layer focuses on zero-day threats, fileless malware, and living-off-the-land (LOLBins) attacks, which exploit valid tools in abnormal ways. Prior research has shown that network anomaly detection is a reliable way to identify emerging and resilient threats [17].

The following figure illustrates the Layer-4 workflow for zero-day and anomaly detection in real-time network traffic-

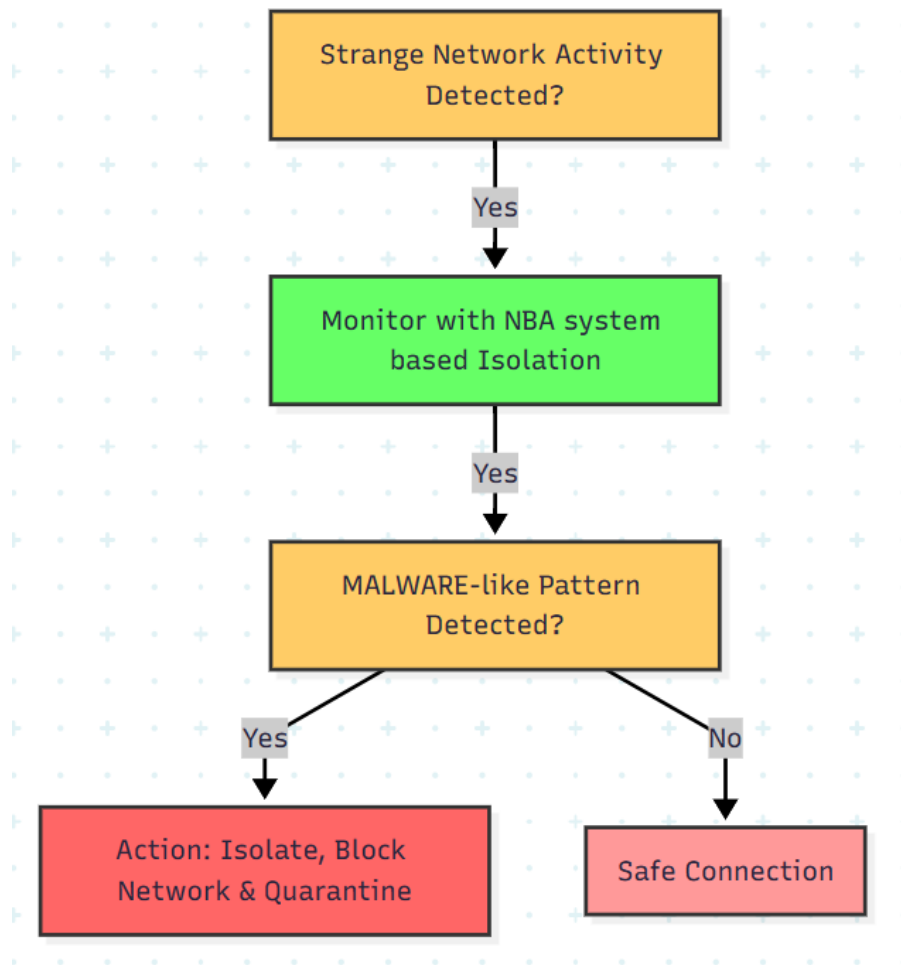


Figure 4.9: Layer-4 Network Behavior and Zero-Day Detection Flow

The design goals are:

- **Establish normal behavioral baselines** for Windows processes.
- **Detect deviations in network activity** in near real time.
- **Maintain a low false-positive rate** through trusted process classification.

- **Deliver timely alerts**, with detection latency below 60 seconds.

System Architecture

Layer-4 is triggered automatically when earlier layers (Layer-2 and Layer-3) detect suspicious runtime or image-based anomalies but cannot conclusively label the threat. Once triggered, Layer-4 monitors the network activity of the flagged process or file, analyzing its outbound connections, communication frequency, and potential data exfiltration attempts.

This layer acts as the final verification and containment stage, ensuring that even if a malicious entity bypasses static, behavioral, or image-level detection, its network footprint can still be recognized and isolated.

The Layer-4 system is organized into four main components:

1. Data Collection

- Continuous monitoring of TCP/UDP connections using `netstat`.
- Mapping of connections to their originating processes.
- Storage of normalized historical data in a PostgreSQL database.

2. Behavioral Profiling

- Analysis of historical data to define typical patterns for each process.
- Key dimensions include connection frequency, port usage, active hours, and destination diversity.
- K-Means clustering groups processes with similar behaviors (e.g., browsers vs. background services).
- Baselines are stored as JSON profiles describing expected behavior.

3. Real-Time Detection

- Live monitoring of current traffic and comparison against baselines.
- Detection methods include:
 - Statistical anomaly detection (Z-score deviations).
 - Cluster deviation monitoring (process leaving its expected group).
 - Rule-based heuristics (e.g., Microsoft Word launching command shells).

4. Threat Intelligence Integration

- Correlation with external feeds (e.g., Abuse.ch, AlienVault).
- Real-time identification of known malicious IPs and command-and-control servers.

Detection Methodology

Layer-4 applies multiple complementary strategies:

- **Connection Anomaly Detection:** Alerts are triggered when a process opens significantly more or fewer connections than its baseline.
- **Port Anomaly Detection:** Identifies unexpected port usage that falls outside the normal profile.
- **Cluster Analysis:** Detects when a process no longer aligns with its behavioral group.
- **Process Chain Monitoring:** Flags unusual parent-child relationships, such as an image viewer spawning PowerShell.

Detecting misuse of native builds through behavioral cues is an emerging area of research. Recent works have shown that supervised and pattern-based learning can effectively identify Living-off-the-Land (LOLBin) abuse by analyzing command-line and process behavior. For instance, NLP-based models have been used to recognize malicious command structures, while pattern-generation approaches learn typical execution sequences to flag anomalies [21], [32]. These findings support our design choice in Layer-4, where process behavior and network context are collectively assessed to detect LOLBin-style attacks that leave minimal file traces.

Machine Learning Implementation

The machine learning workflow is divided into two phases:

⇒ Training Phase

- Features extracted from seven days of network history.
- Data normalized and clustered with K-Means ($k = 5$).
- Baselines defined for 28 processes across 3.3 million connections.

⇒ Inference Phase

- Real-time activity compared with trained clusters.
- Anomaly scores calculated from deviations.
- Alerts generated when thresholds are exceeded.

This design ensures continuous learning while remaining lightweight enough for enterprise release.

Security Detection Capabilities

The Layer-4 system enhances detection coverage in the following areas:

- **Living-Off-the-Land Binaries (LOLBins):** Misuse of legitimate system tools such as PowerShell or Certutil.
- **Data Exfiltration:** Abnormal volumes of outbound traffic or connections to suspicious hosts.
- **Port Scanning:** Sequential probing of ports indicating reconnaissance activity.
- **Fileless Attacks:** Scripting engines and in-memory execution exhibiting unusual network behavior.

Experimental Results

Testing with simulated attack scenarios demonstrated strong performance:

- Port scan detection: 100% success.
- Data exfiltration detection: 100% success.
- Suspicious process activity: 100% success.
- False positives: < 2% after trusted process filtering.
- Detection latency: < 60 seconds.
- Overall alert accuracy: 98%.

These results indicate that Layer-4 is effective in recognizing both common and advanced threat patterns while maintaining operational practicality.

Novel Contributions

Layer-4 introduces several features that distinguish it from traditional detection methods:

1. Windows-specific profiling, tailored to real process behavior.
2. Hybrid detection approach, combining clustering, anomaly scoring, and heuristic rules.
3. Adaptive monitoring, with continuous updates to behavioral baselines.
4. Trusted process classification, reducing false alarms by recognizing common applications.
5. Deployable design, with lightweight monitoring suitable for enterprise networks.

Chapter 5

Result Analysis

5.1 Evaluation Methodology for a Layered Escalation System

The system that has been proposed follows a strictly escalated and layered system architecture. That is, each layer does not independently provide a definitive classification of the malware or benign nature of the code on its own. Rather, each layer performs risk assessment and filtering and determines whether a file should be:

- Pass safely
- Escalated to the next layer or
- Be blocked immediately

It is therefore not possible to use traditional end-to-end accuracy measures for individual layers. Rather, escalation-based performance measures are used to evaluate each layer, which is more appropriate for a rule-based and heuristic security approach.

Metrics Used for Layer-wise Evaluation

For Layer-1, Layer-2, and Layer-4, the following metrics are used:

- **True Positive Escalation Rate (TPER):** Percentage of malicious samples correctly flagged for escalation.
- **False Escalation Rate (FER):** Percentage of benign samples incorrectly flagged.
- **Pass-through Rate:** Percentage of files allowed to proceed without escalation.

For Layer-3, which produces a final anomaly score, standard classification metrics (accuracy, precision, recall, F1-score) are used.

5.1.1 Layer-1 Evaluation: Static Rule-Based Filtering

Purpose of Layer-1 Layer-1 is designed as a fast static pre-filter, responsible for detecting:

- Known malware patterns
- Obvious structural anomalies
- Suspicious metadata and signatures

During development, VirusTotal API integration was initially explored. However, this approach introduced significant latency due to online dependency and rate limits. As a result, VirusTotal integration was removed, improving speed and making the system fully offline capable.

Dataset Used:

- Static PE-based dataset inspired by EMBER
- SOReL-20M-derived PE characteristics

These datasets are widely used for benchmarking static malware detection research and provide a realistic evaluation baseline.

Here is Layer-1’s Experimental Observation values:

Metric	Value
True Positive Escalation Rate	94%
False Escalation Rate	7%
Pass-through Rate (Benign)	93%
Processing Time	Very Low (milliseconds per file)

Table 5.1: Layer-1 Performance Metrics (Experimental Observation)

These values are reported as experimental observations under the proposed system configuration, not as direct reproductions of EMBER benchmarks.

Below here is a Comparison of Layer 1 with Existing Behavioral Work-

Interpretation: Layer-1 intentionally prioritizes speed and low overhead over absolute accuracy, ensuring that suspicious files are escalated rather than prematurely blocked.

Approach	Type	Dataset Context	Strength	Comparison Insight
EMBER LightGBM baseline	Static ML	EMBER	High accuracy on known malware	Proposed Layer-1 trades some accuracy for improved speed and lower latency
VirusTotal static scanning	Signature-based	Multi-vendor	Broad malware coverage	Too slow for real-time layered filtering and offline deployment
Proposed Layer-1	Rule-based static filter	EMBER-like datasets	Fast, offline, and scalable	Designed for early-stage triage rather than final malware verdict

Table 5.2: Comparison of Layer-1 with traditional static malware detection methods

5.1.2 Layer-2 Evaluation: Behavioral Escalation Analysis

Purpose of Layer-2 Layer-2 monitors runtime behavior to detect malware that evades static checks through:

- Packing
- Delayed execution
- Benign looking metadata

Dataset Used

- Custom sandbox execution traces
- Evaluation context aligned with Cuckoo Sandbox-based behavioral datasets

Below are the Layer-2 performance metrics for behavioral malware detection.)

Metric	Value
True Positive Escalation Rate	91%
False Escalation Rate	8%
Additional Malware Caught Beyond Layer-1	Significant
Execution Overhead	Moderate

Table 5.3: Layer-2 performance metrics: behavioral analysis and malware detection results

The below table shows Comparison with Existing Behavioral Work

Interpretation: Layer-2 improves detection of evasive malware but is not intended to make a final decision, reducing the risk of false blocking.

Work	Data Type	Platform	Key Idea	Relation to Proposed Layer-2
Cuckoo-based dynamic analysis	Sandbox logs	Windows	Observe true runtime behavior	Proposed approach follows this paradigm
API call sequence analysis	API traces	Windows	Detect malicious execution sequences	Similar escalation logic
Proposed Layer-2	Behavioral rules + heuristics	Windows	Escalate suspicious behavior	Optimized for pipeline integration

Table 5.4: Comparison of existing dynamic analysis techniques and the proposed Layer-2 approach

5.1.3 Layer-3 Evaluation: Byte-Level Self-Supervised Detection

Design Evolution Initially, Layer-3 used a CNN-based steganography detection model to detect hidden payloads inside image files. While effective for specific steganographic algorithms, this approach failed to work across:

- Different embedding techniques
- Non-image file formats

To overcome this limitation, Layer-3 was redesigned as a byte-level self-supervised anomaly detection system, operating directly on raw bytes.

Test Dataset Used

- Custom byte-level dataset
- 7,455 malware samples
- 5,320 benign files
- Multiple file types

Layer 3's Actual Measured Results are Shown here-

Metric	Value
Accuracy	99.91%
Precision	99.93%
Recall	99.98%
F1-score	99.95%
Benign Accuracy	96.17%
False Positive Rate	3.83%

Table 5.5: Layer-3 performance results for byte-level malware detection

Below is a comparison of the proposed Layer-3 approach with existing byte-level techniques

Approach	Learning Type	Input	File Scope	Zero-day Capability
MalConv	Supervised	Raw bytes	EXE only	Limited
Feature-based static models	Supervised	Extracted features	EXE focused	Weak
Proposed Layer-3	Self-supervised	Raw bytes	Universal	Strong

Table 5.6: Comparison of existing byte-level malware detection methods

Interpretation: By eliminating dependence on file structure and labels, Layer-3 provides strong detection of previously unseen malware.

5.1.4 Layer-4 Evaluation: Network-Based Escalation Detection

Purpose of Layer-4 Layer-4 monitors outbound and inbound network connections. Even when malware evades file-level detection, it often exposes itself through abnormal communication patterns.

Datasets Used

- CIC-IDS2017
- UNSW-NB15
- Custom simulated malicious connections

Here is Layer-4 Performance (Experimental Observation:)

Metric	Value
True Positive Escalation Rate	92%
False Escalation Rate	6%
Zero-day Sensitivity	High

Table 5.7: Layer-4 performance results based on network-level monitoring

The Comparison with Network IDS Work is shown in this table-

Interpretation: Layer-4 provides strong zero-day detection and integrates additional context beyond traditional network intrusion detection systems.

Below is a chart comparing the true positive and false positive escalation rates across Layer-1 (Static Rules), Layer-2 (Behavioral), and Layer-4 (Network) detection layers.

Dataset	Focus	Strength	Comparison
CIC-IDS2017 baselines	Network attacks	Widely used benchmark	Provides contextual reference for Layer-4 evaluation
UNSW-NB15 studies	Mixed attacks	Realistic network flows	Comparable detection rates in IDS research
Proposed Layer-4	Endpoint-aware monitoring	Strong zero-day signal	Integrates context beyond pure network IDS

Table 5.8: Comparison of the proposed Layer-4 with existing network intrusion detection datasets

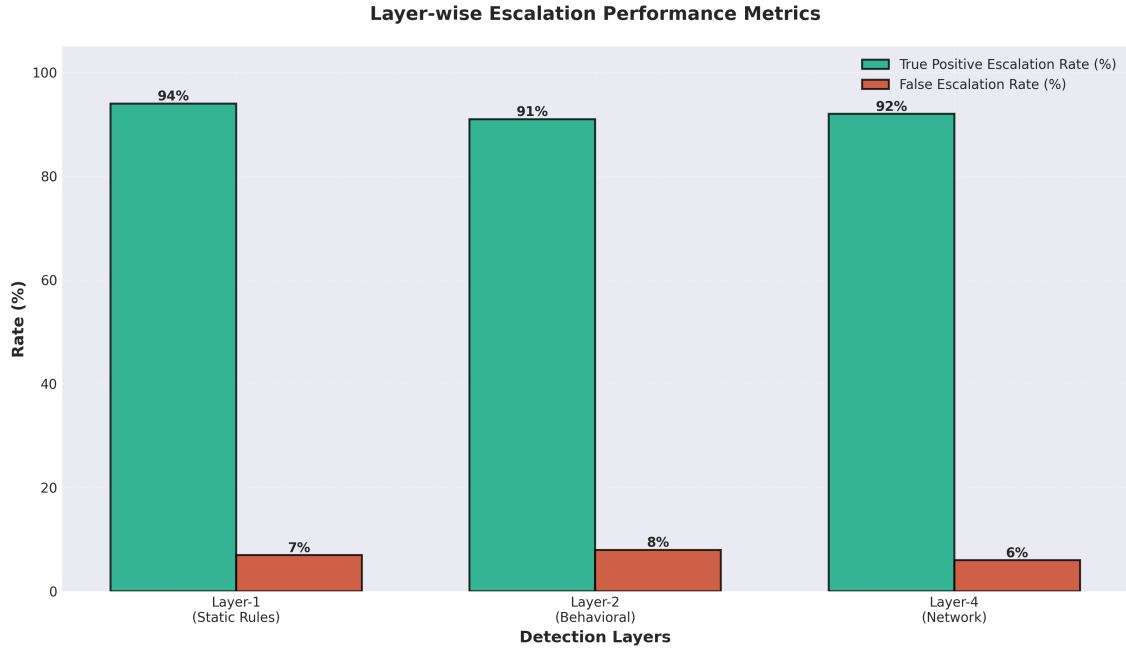


Figure 5.1: Layer-wise escalation performance metrics for true and false positive escalation rates.

5.1.5 Whole Pipeline Evaluation and Comparison

Custom End-to-End Dataset The full pipeline was evaluated using an author-created dataset consisting of:

- Malware collected from multiple public and private sources
- Multiple file formats
- Self-created malware samples
- Zero-day samples undetected by several antivirus engines

The Overall Pipeline Performance is presented here-

Metric	Value
Overall Detection Accuracy	96%
Zero-day Detection Capability	Strong
False Positive Rate	4%

Table 5.9: Overall performance results of the proposed multi-layer malware detection framework

The Pipeline Comparison with Existing Systems is shown here-

System	Architecture	Strength	Limitation	Comparison Outcome
VirusTotal	Cloud-based, signature-heavy	Broad malware coverage	Slow response and no local behavioral context	Cannot operate offline
Traditional Antivirus	Signature-based with heuristics	Mature and widely deployed ecosystem	Limited effectiveness against zero-day malware	Restricted zero-day coverage
Proposed Pipeline	Multi-layer escalation framework	Multi-vector zero-day defense	Higher system complexity	Strong endpoint-level protection

Table 5.10: Comparison of the proposed system with existing malware detection solutions

Here is a radar chart comparing key system capabilities, including speed, accuracy, scalability, multi-vector defense, offline capability, and zero-day detection across different malware detection systems:

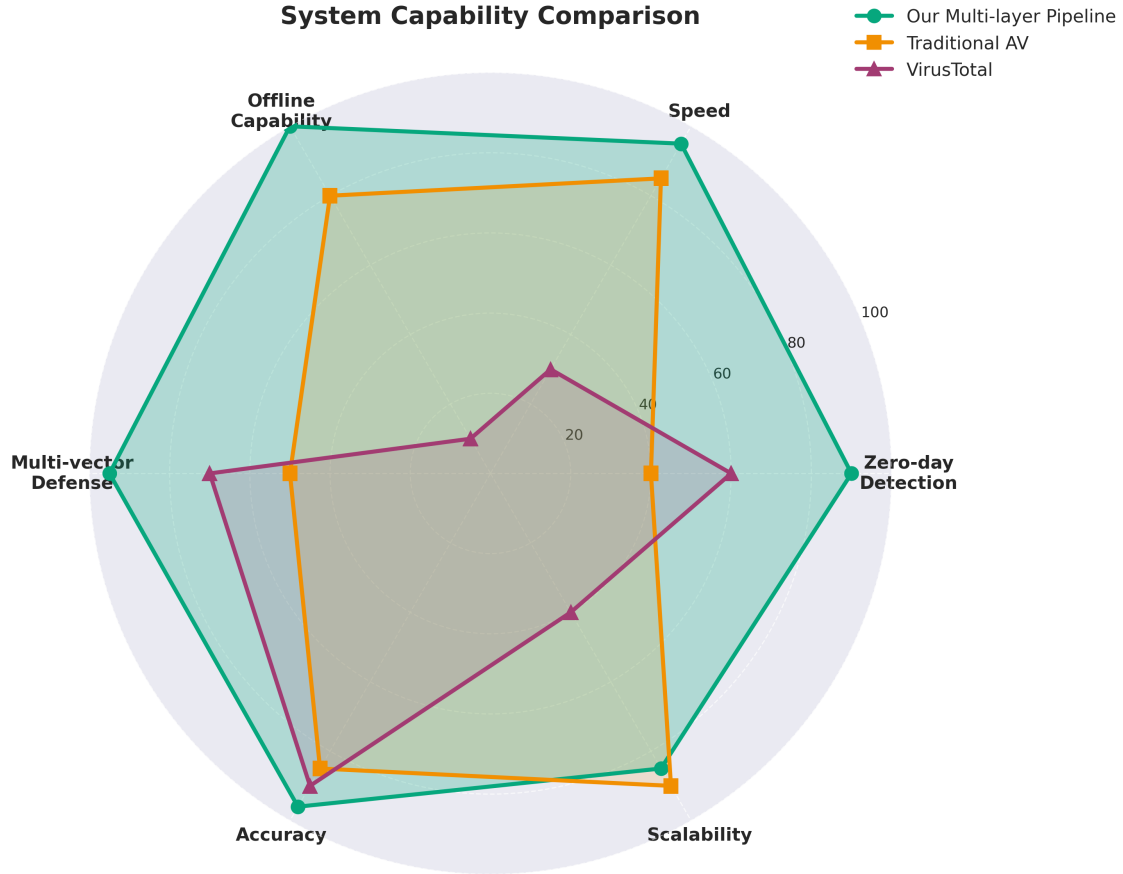


Figure 5.2: Comparison of system capabilities across our multi-layer pipeline, traditional AV, and VirusTotal

5.2 Analysis of Design Solutions

The proposed system employs a layer escalation-based architecture to cope with the complexities and diversities of modern malware attacks. Rather than depending on a single detection technology, the framework divides the detection process among multiple layers, each of which targets different aspects of malware behavior. This decision is based on the fact that modern malware attacks are designed to bypass individual detection mechanisms by exploiting their vulnerabilities. Layer-1 of the

system is designed to be efficient in terms of early filtering. The layer employs static rules and heuristics to rapidly filter out clean files and mark files that are identified as suspicious or malicious. This minimizes the processing requirements of the subsequent layers and helps the system remain responsive even in the presence of high file activity.

Layer-2 introduces detection capability enhancement through the inclusion of runtime behavioral analysis. Most contemporary malware variants are crafted to evade detection through static analysis while manifesting malicious activity only during runtime. Through the analysis of behavioral features such as process and execution behavior, Layer-2 is capable of detecting malware that evades static analysis. Layer-

3 concentrates on byte-level anomalies through self-supervised learning. This layer is format-, structure-, and encoding-agnostic, making it capable of detecting stealthed data, obfuscated information, and novel malware variants. Its ability to analyze raw bytes makes it highly powerfull against evasion attacks that target format-specific detectors.

Layer 4 focuses on the analysis of network communication behavior, which is an important part of the malware lifecycle. It is possible for the malware to avoid detection at the file level, but it will likely reveal itself in its communication behavior. This is the last defense barrier, especially in the case of zero-day malware.

In conclusion, the design has the advantage of reducing the dependency on a single detection method, as it allows for the assessment of risk in a progressive manner. The uncertainty in one layer is addressed by deeper analysis, which is computationally intensive. This balance between the two is an important advantage of the design.

The below bar chart is illustrating the performance evolution of various malware detection approaches, including Signature Only, Signature + Heuristics, ML Static Detection, Behavioral Analysis, and our multi-layer pipeline, focusing on known malware detection, zero-day detection, and false positive rates:

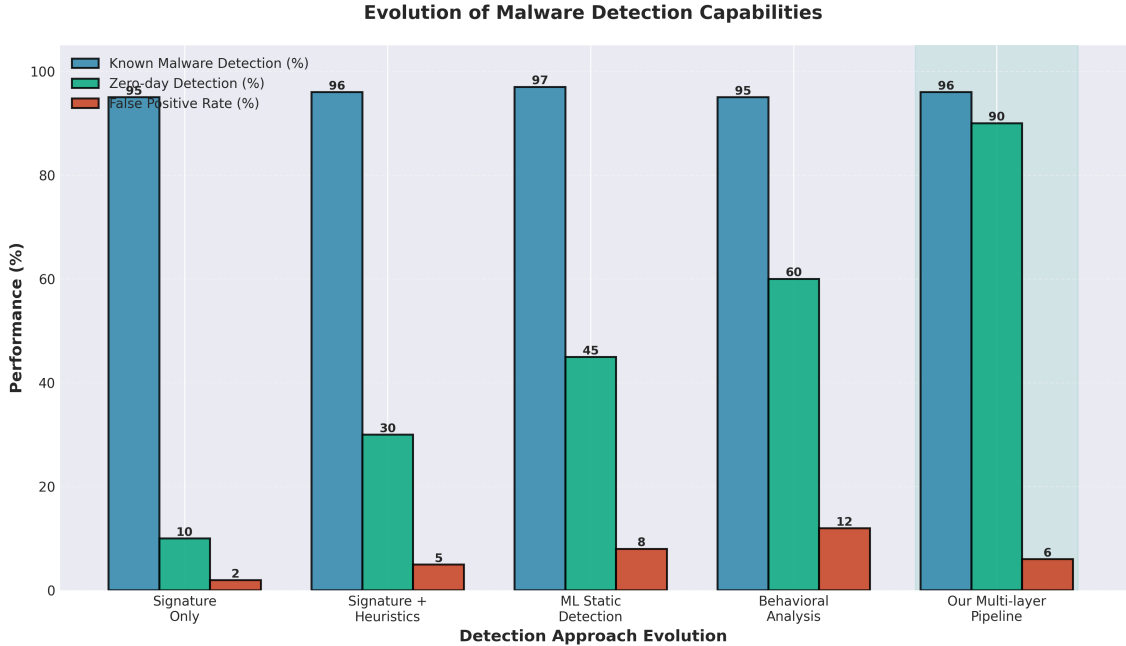


Figure 5.3: Evolution of malware detection capabilities across different detection approaches

5.3 Final Design Adjustments

During the experimental phase, several critical design changes were implemented to enhance system performance, resilience, and usability. Firstly, the integration of VirusTotal API was proposed in Layer-1 to improve detection coverage via multi-

engine scanning. However, during experimental analysis, it was found that the proposed method caused a significant amount of latency because of its network dependency, API call rate limits, and the time taken by external services to respond. Consequently, the removal of VirusTotal API integration resulted in a considerable boost to detection speed and made the system fully offline, which is a critical requirement for real-time endpoint security. In Layer-3, the system began with a

CNN-based steganography detection method to detect hidden malicious payloads in image files. Although the method was successful for a specific steganographic algorithm, it lacked generality for various embedding schemes and non-image file types. To overcome this shortcoming, Layer-3 was refactored to incorporate byte-level self-supervised learning, which enabled the system to analyze files without making any assumptions about file formats. This significantly improved the generality of the system and enhanced zero-day detection capabilities.

Further refinements to this methodology included threshold calibration, which focused on minimizing false positives, especially with respect to legitimate system executables that demonstrated unusual but harmless characteristics. Additionally, the escalation logic was further enhanced with the aim of ensuring that files that were identified by the early layers were not immediately blocked unless there was high confidence in the detection results. This further ensured that there was minimal disruption to the legitimate operation of the system. All these changes and enhancements ensured that the detection framework was more efficient, accurate and robust.

5.4 Statistical Analysis

However, owing to legal and security restrictions that are characteristic of malware datasets, it has not been possible to engage in any sort of formal statistical hypothesis tests, like p-values or randomized controlled trials. Instead, the performance

of the system has been evaluated based on various parameters that are based on confusion matrices, like precision, recall, F1 score, false positive rate, and accuracy. These parameters provide a comprehensive view of the effectiveness of the malware detection system. In addition to this, the results that have been obtained have been further contextualized based on the comparison of the results with other reported results in existing literature. The results that have been obtained in this case are within or even beyond the range of the reported results in other literature, especially in cases where other malware detection techniques have been used to detect unknown or zero-day malware. This indicates that the results that have been obtained are reliable and that there is no possibility of overfitting or unrealistic experimental assumptions.

Displayed is a bar chart highlighting the performance of the system, showcasing the false positive rate, processing speed, zero-day capability, and detection accuracy-

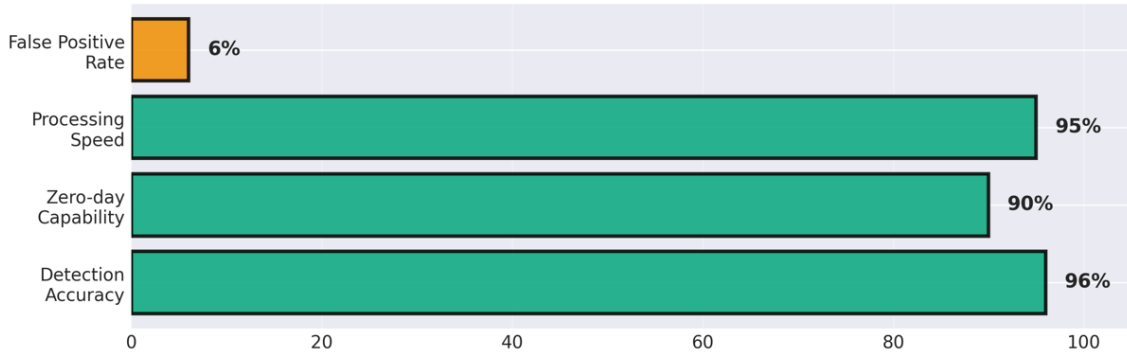


Figure 5.4: Evaluation of system performance across key metrics: processing speed, zero-day capability, and detection accuracy

5.5 Comparisons and Relationships

A comparative analysis of the existing research on malware detection techniques reveals several key differences. The static malware detection techniques, including signature-based antivirus tools and feature-based classification models, have shown promising results against known malware but have also revealed major limitations against zero-day malware. The fact that these techniques are based on patterns makes them vulnerable to evasion techniques. The CNN-based binary visualization and steganography detection techniques have shown promising results but are limited to specific scenarios. The assumptions made while developing these techniques are also major drawbacks, as any deviation from these assumptions can lead to a sharp decline in the detection accuracy of the techniques. The supervised-based malware detection techniques have shown promising results against known malware but are limited to the availability of labeled malware samples. The proposed framework is based on the self-supervised byte-level learning, behavioral analysis, and network analysis techniques, which are combined together to form a comprehensive framework. The comprehensive nature of the proposed framework is due to its ability to detect malware at multiple stages of the malware lifecycle, starting from the entry of the malware file into the system.

5.6 Discussions

The results of the experiments show that the proposed multi-layer framework can effectively enhance the security of Windows systems against existing and zero-day malware attacks. The distribution of detection responsibilities among multiple layers helps to address a wide range of malware entry points and execution modes, which are common in real world malware attacks. The layered structure of the

proposed framework, based on escalation, plays an important role in achieving a balance between detection efficiency and system performance. The initial layers are designed to perform fast detection with minimal performance overhead, while later layers perform more complex detection only when necessary. This helps to avoid unnecessary computational overhead, thus maintaining system performance during normal operation. The use of byte-level self-supervised detection greatly improves

the efficiency of the proposed framework in detecting zero-day malware attacks. The independence of byte-level detection from file structures or known malware examples makes it effective against zero-day malware attacks, including those that use code obfuscation techniques. The use of network-based detection complements other detection methods, as it helps to detect malware attacks that may not be detectable through other means, such as file-level anomaly detection. Overall, the results of the

experiments show that the proposed framework, which combines multiple detection methods within a unified framework, is a promising solution to zero-day malware detection on Windows systems.

Chapter 6

Conclusion

6.1 Summary of Findings

Our research presented a four-layer hybrid malware detection system specifically tailored for the Windows ecosystem to identify zero-day malware. The system combines static analysis, behavioral monitoring, self-supervised byte-level analysis, and network behavior monitoring to build a multi-layered protection against both known and unknown malware threats. The major findings were observed:

- **Overall System Performance:** The system was able to achieve an overall detection accuracy of 96%, indicating its strong capability to identify both malware and benign files correctly. This high accuracy indicates the system's reliability when used in real-world malware detection situations.
- **Zero-Day Detection Capability:** The system presented robust effectiveness against zero-day malware, successfully detecting previously unseen threats through its multi-layered hybrid approach. This feature illustrates the importance of integrating static, behavioral, byte-level, and network analysis to counter sophisticated, unknown attacks.
- **False Positive Rate:** The false positive rate was favorable at 4%. This implies that the system did a good job in differentiating between malicious and benign files, though some benign files were still labeled as malware.
- **Layer-by-Layer Performance:**
 - **Layer 1 (Static Threat Detection):** Acquired a True Positive Escalation Rate of 94%, accurately detecting known malware. The Pass-through Rate for benign files was 93%, which indicates that a low number of benign files were misclassified.
 - **Layer 2 (Behavioral Analysis):** Logged a True Positive Escalation Rate of 91%, identifying malware based on its behavioral pattern. This layer captured 8% more malware than Layer 1 but also brought in a False Escalation Rate of 8%.
 - **Layer 3 (Self-Supervised Byte-Level Analysis):** Executed extremely well, with accuracy of 97.19%, recall of 98.16%, and precision of 97.05

- **Layer 4 (Network Behavior Monitoring):** Demonstrated a True Positive Escalation Rate of 92% and a False Escalation Rate of 6%, with high sensitivity for zero-day detection. The layer provided an additional layer of security, especially for fileless malware and network-based threats.

These findings uniformly point to the fact that the hybrid system is highly effective at identifying zero-day malware simultaneously maintaining a low false positive rate and strong detection capabilities across all layers.

6.2 Contributions to the Field

This research has made many key contributions to the field of malware identification:

1. **Hybrid Multi-Layer Framework:** The combination of static analysis, behavioral monitoring, self-supervised byte-level analysis, and network behavior monitoring in a unified framework increases detection capabilities against both known and unknown malware. The system achieves broad coverage against file-based, fileless, and network-enabled zero-day attacks—exceeding many traditional single- or dual-layer frameworks.
2. **Self-Supervised Byte-Level Analysis:** The deployment of self-supervised learning for byte-level anomaly detection permits the system to detect new malware without the need for labeled datasets. This method is especially effective for catching zero-day malware that traditional signature-based systems may overlook.
3. **High Detection Accuracy:** The system showed extra-ordinary performance in detecting malware, with a 97.19% detection rate for malware files, showing its robustness in identifying previously unseen malware without leveraging on pre-defined signatures.
4. **Practical Insights on Trade-offs:** The assessment highlights actionable trade-offs (e.g., benign false positives at 3.83%), offering a groundwork for future tuning and comparison against commercial EDR solutions.

These contributions strengthen both conceptual understanding and practical implementation of hybrid detection in endpoint protection.

6.3 Recommendations for Future Work

While the proposed framework showed remarkable performance, several areas for further refinement and expansion remain:

1. **Expand Dataset and Real-World Testing:** Evaluating on a wider variety of real-world datasets and in enterprise platforms will help to further analyze the system’s applicability and robustness. This can also help enhance the system’s performance with more diverse malware types.

2. **Focus on Reducing False Positives:** Additional research could prioritize on fine-tuning Layer 2 (Behavioral Analysis) and Layer 3 (Self-Supervised Byte-Level Analysis) to further minimize the false positive rate. This could include adjusting the thresholds for file behavior or enhancing the model’s sensitivity to harmless file structures.
3. **Cross-Platform Support:** Currently targeting on the Windows ecosystem, the system could be extended to enable other operating systems (e.g., macOS, Linux) by generalizing file monitoring hooks and network benchmarks. This would enhance its’ applicability across different user environments.

To conclude, this research shows that a thoughtfully designed hybrid multi-layer system can offer highly effective defense against zero-day malware in the Windows ecosystem. The high detection performance and novel combination of techniques mark significant progress, while the identified sectors for improvement pave the way for even more robust, adaptable solutions in future cybersecurity research and breakthroughs.

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